

25/04/2025

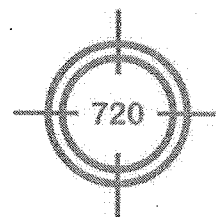


Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced INTENSIVE Mastery for 720)

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MM : 720

CST-II

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Biology, Physics and Chemistry having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



BIOLOGY

1. Which of the following is **odd** w.r.t. the roles performed by ethylene as a PGR?
 - (1) Breaking seed and bud dormancy
 - (2) Promoting root growth and root hair formation
 - (3) Promotes male flowers in mango
 - (4) Accelerating abscission in flowers and fruits
2. Cells of meristematic phase do **not** show
 - (1) Presence of dense cytoplasm
 - (2) Increased vacuolation
 - (3) High respiratory rate
 - (4) A large conspicuous nucleus
3. Which of the following cells are polyploid?
 - (1) Microspore mother cells
 - (2) Tapetal cells
 - (3) Generative cells of pollen grains
 - (4) Cells of endothecium of anther wall
4. Read the statements given below and identify the **correct** option.
Statement – I : A fruit which develops from any floral part except ovary is said to be a false fruit.
Statement – II : Some species of Asteraceae family have evolved the mechanism of apomixis.
 - (1) Both the statement-I and statement-II are true
 - (2) Both the statement-I and statement-II are false
 - (3) Statement-I is false but statement-II is true
 - (4) Statement-I is true but statement-II is false
5. Read the following Assertion (A) and Reason (R) statements and choose the **correct** option from the following.
Assertion (A) : A typical angiosperm embryo sac at maturity, though 8-nucleate is 7-celled.
Reason (R) : In the embryo sac-egg apparatus consists of two synergids and one egg cell, and three antipodals are present towards the chalazal end.
 - (1) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (2) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (3) (A) is true but (R) is false
 - (4) (A) is false but (R) is true
6. Myotonic dystrophy and cystic fibrosis both are
 - (1) Autosomal disorders
 - (2) Recessive gene disorders
 - (3) X-linked recessive disorders
 - (4) X-linked dominant disorders
7. Choose the **correct** option w.r.t. sickle-cell anaemia.
 - (1) It is the example of frame-shift mutation.
 - (2) It is transmitted from a healthy homozygous father and a carrier mother to the male child.
 - (3) Homozygous individuals for Hb^S show the diseased phenotype.
 - (4) It is caused by the substitution of Valine by Glutamic acid at the sixth position of the beta globin chain of haemoglobin.
8. Four genes a, b, c and d are present on a chromosome. Percentage of recombination between a and b = 20%, a and d = 5%, b and c = 10% and b and d = 15%. Which of the following can be the sequence of genes on the chromosome?
 - (1) adcb
 - (2) cdab
 - (3) acbd
 - (4) dcab
9. Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.
Assertion (A) : Dominance is an autonomous feature of a gene.
Reason (R) : Dominance depends much on the gene product and the production of a particular phenotype from this product.
 - (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true but (R) is false
 - (4) (A) is false but (R) is true
10. Which of the following steps is **not** involved in hnRNA processing in an eukaryotic cell?
 - (1) Polyadenylation
 - (2) Transcription
 - (3) Capping
 - (4) Splicing

11. If the sequence of codon in mRNA is 5' AGU 3', then what would be the sequence of anticodon on its corresponding tRNA?
- 3' CAU 5'
 - 5' GUA 3'
 - 5' ACU 3'
 - 5' UCA 3'
12. During the process of DNA transcription, a promoter is
- A region that codes for an enzyme initiating the transcription
 - A sequence of DNA that terminates transcription
 - A sequence of DNA where RNA polymerase binds to initiate transcription
 - A protein that assists the unwinding of DNA helix
13. Read the given statement and choose the **correct** option to fill in the blank.
lac operon may function constitutively if a mutation occurs in the _____ region.
- Operator
 - lac z*
 - Inducer
 - lac a*
14. In which of the following parts of plant, would you expect to find the conjunctive tissue and pericycle?
- Leaves of monocots
 - Leaves of dicots
 - Stems of grasses
 - Roots of dicots
15. Read the following characteristics.
- Presence of casparian strips in endodermis
 - Absence of bundle sheath
 - Ground tissue differentiated into cortex and pith
 - Medullary rays found in between vascular bundles
 - Epidermis covered with thin layer of cuticle
- Which of the given characteristics are **true** for dicot stem?
- (b) and (e) only
 - (a), (c) and (e) only
 - (c) and (d) only
 - All except (a)
16. All of the following are components of heavy meromyosin, **except**
- Actin binding site
 - ATP binding site
 - Cross arm
 - Tropomyosin
17. How many cycles of PCR are required to generate 256 copies of DNA from a 10 kb dsDNA molecule?
- 6
 - 10
 - 8
 - 12
18. Choose the **incorrect** match.
- β -galactosidase – Reporter enzyme
 - Kanamycin encoding gene – Selectable marker
 - Molecular scissors – Present in prokaryotes
 - EcoRI* - Produces blunt ends
19. Which of the following pairs of connective tissues lack fibres of structural proteins?
- Areolar tissue, Adipose tissue
 - Bone, Cartilage
 - Dense regular connective tissue, Dense irregular connective tissue
 - Blood, Lymph
20. In humans, the brain stem is formed by
- Midbrain, cerebellum and pons
 - Thalamus, amygdala and hippocampus
 - Pons, midbrain and medulla oblongata
 - Hypothalamus, hindbrain and cerebrum
21. Choose the **incorrect** statement.
- Agarose gel electrophoresis is employed to check the progression of a restriction enzyme digestion.
 - The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with the native plasmid of *Meloidogyne incognita*.
 - In PCR, a certain enzyme extends the primers using the nucleotides provided in the reaction and the genomic DNA as template.
 - Digestion by restriction enzyme is performed by incubating purified DNA molecules with the restriction enzyme at the optimal conditions.
22. Among the following, choose the property which is common between functional unit of neural system and anatomical unit of muscle.
- Contractility
 - Elasticity
 - Extensibility
 - Excitability

23. Match the column I with column II

	Column I		Column II
a.	Sino-atrial node	(i)	Thick fibrous tissue
b.	Atrio-ventricular node	(ii)	Thin muscular wall
c.	Inter-atrial septum	(iii)	Lower left corner of right atrium
d.	Atrio-ventricular septum	(iv)	Right upper corner of right atrium

Choose the correct option.

- (1) a(iv), b(i), c(ii), d(iii)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(ii), b(iii), c(i), d(iv)

24. Read the following carefully w.r.t. an adult healthy human.

- a. Intercellular matrix is hard, non-pliable and resists compression
 - b. Not found between the adjacent lumbar vertebrae
 - c. Cells of this tissue are enclosed in small cavities within the matrix secreted by them.
- Choose the option which represents only the correct feature(s) w.r.t. cartilages.

- (1) a, b and c
- (2) only a and b
- (3) only c
- (4) a and c only

25. All of the following exhibit complete double circulation, except

- (1) *Columba*
- (2) *Pavo*
- (3) *Ornithorhynchus*
- (4) *Carcharodon*

26. Intercalary meristem differs from lateral meristem as the former is

- (1) Found at the apices of stem
- (2) Present in radial directions to increase the girth of plant parts
- (3) Involved in the activity that adds to the length of the plant or its organs
- (4) Generally not present in plants from the very beginning of its life

27. Select the correctly matched pair.

- (1) Nucleoid – packaged structure of eukaryotic DNA
- (2) UGG – non-degenerate codon
- (3) Looks like a clover-leaf – tertiary structure of tRNA
- (4) Regulator gene of *lac* operon – does not transcribe mRNA for protein synthesis

28. Middle lamella between the plant cells is chiefly made up of

- (1) Chitin
- (2) Calcium pectate
- (3) Cellulose
- (4) Peptidoglycan

29. Match the following columns and choose the correct option.

	Column – I		Column – II
(a)	Plasmodesmata	(i)	Extract energy from carbohydrates to form ATP
(b)	Mitochondria	(ii)	Site for rRNA synthesis
(c)	Axoneme	(iii)	Cytoplasmic connections
(d)	Nucleolus	(iv)	Core of cilia and flagella

- (1) (a-i), (b-iii), (c-ii), (d-iv)
- (2) (a-iii), (b-i), (c-iv), (d-ii)
- (3) (a-iii), (b-i), (c-ii), (d-iv)
- (4) (a-iii), (b-ii), (c-iv), (d-i)

30. In which of the following phases, there is no duplication of the cell organelles in animal cells?

- (1) G₁ Phase
- (2) S phase
- (3) M phase
- (4) Interkinesis

31. Find the incorrectly matched pair w.r.t. cell division.

- (1) Quiescent stage – G₀ phase
- (2) Crossing over – Exchange of genetic material between non-sister chromatids of homologous chromosomes
- (3) Intrameiotic interphase – Metabolic stage between Telophase I and Prophase II
- (4) Telophase I – Results in the formation of tetrad of cells

32. Read the following statements and choose the incorrect one w.r.t. microfilaments.

- (1) They are solid, unbranched, rod-like fibrils
- (2) They are mainly composed of a globular protein, actin and also have filamentous protein myosin
- (3) They are involved in cytoplasmic streaming and amoeboid movements
- (4) They are involved in formation of scaffolds for chromatin

33. Read the following statements of Assertion (A) and Reason (R).

Assertion (A) : Meiosis helps in maintaining the cell size.

Reason (R) : An overgrown somatic cell is induced to divide so that it maintains a proper surface area to volume ratio.

In light of the above statements, choose the **correct** answer from the following options.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

34. Read the following statements and select the **incorrect** ones for the members of algae that reproduce asexually by zoospores.

- (a) They occur only at great depth in oceans.
- (b) The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.
- (c) The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin.
- (d) The shape of chloroplast may be discoid, plate-like, reticulate or cup-shaped in different species.

- (1) (a) and (b) only
- (2) (c) and (d) only
- (3) (a) and (d) only
- (4) (b) and (c) only

35. *Marchantia* differs from *Funaria* as in the former

- (1) The plant body is a leafy gametophyte which has multicellular and branched rhizoids
- (2) Spores are formed after meiosis and have an elaborate mechanism for their dispersal
- (3) Each archegonium produces a fertile egg
- (4) Male and female sex organs are developed on different thalli

36. Microorganisms such as *Lactobacillus* are used for the commercial production of

- (1) Acetic acid
- (2) Butyric acid
- (3) Lactic acid
- (4) Citric acid

37. Biological control methods adopted in agriculture pest control are based on the ability of the predator to

- (1) Completely eradicate the pest population in that ecosystem
- (2) Overexploit the prey population
- (3) Regulate the prey population
- (4) First increase and then decrease the prey population

38. Identify the **incorrect** match w.r.t. sacred groves.

- (1) Chanda and Bastar areas – Gujarat
- (2) Aravalli Hills – Rajasthan
- (3) Western Ghat regions – Karnataka
- (4) Khasi and Jaintia Hills – Meghalaya

39. When a species becomes extinct, the plant and animal species associated with it, in an obligatory way, also become extinct. This exemplifies

- (1) Over-exploitation
- (2) Co-extinction
- (3) Alien species invasion
- (4) Fragmentation

40. The plant body is often differentiated into holdfast, frond and stipe in

- (1) *Ulothrix*
- (2) *Spirogyra*
- (3) *Eudorina*
- (4) *Laminaria*

41. In humans, plasma protein which majorly helps in maintaining osmotic balance of body is

- (1) Albumin
- (2) Globulin
- (3) Fibrinogen
- (4) GLUT-4

42. The enzymes that play a very important role in rDNA technology is/are

- (a) Restriction endonuclease
- (b) DNA polymerase
- (c) DNA ligase

Choose the **correct** option.

- (1) (a) only
- (2) (b) and (c) only
- (3) (c) only
- (4) (a), (b) and (c)

43. White muscle fibres differ from red muscle fibres as

- (1) Latter has low amount of myoglobin as compared to former
- (2) Former is fusiform in shape while latter is cylindrical
- (3) Former has high amount of sarcoplasmic reticulum than latter
- (4) Latter only depends on anaerobic process for energy

44. The factors which favour the formation of oxyhaemoglobin are
- (1) High pO_2 , low H^+ concentration, low temperature
 - (2) High pCO_2 , low H^+ concentration, high temperature
 - (3) Low pH, high pO_2 , low H^+ concentration
 - (4) High temperature, low pO_2 , low CO_2
45. Read the given statements and select the **correct** option.
Statement (A): Malonate closely resembles the structure of succinic dehydrogenase and illustrates the example of competitive inhibition.
Statement (B): Competitive inhibitors are often used for the control of bacterial pathogens.
- (1) Both statements (A) and (B) are correct
 - (2) Both statements (A) and (B) are incorrect
 - (3) Only statement (A) is correct
 - (4) Only statement (B) is correct
46. 'X' is a tightly bound co-factor of an enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen. 'X' is also the part of the active site of the enzyme named
- (1) Catalase
 - (2) Carbonic anhydrase
 - (3) Ribozyme
 - (4) Carboxypeptidase
47. All of the following are true w.r.t epithelium that lines the walls of blood vessels, **except**
- (1) Composed of a single layer of flattened cells
 - (2) Involved in the functions like forming a diffusion boundary
 - (3) Free surface possesses microvilli
 - (4) The cells are compactly packed with little intercellular matrix
48. Inorganic catalysts differ from enzymes as
- (1) Former possess a crevice or pocket in their secondary structure into which the substrate fits
 - (2) Latter accelerate the rate of reaction by increasing the activation energy
 - (3) Former work efficiently at high temperatures and high pressures
 - (4) Latter always get inactivated at high temperature
49. Choose the odd one w.r.t structures that possess heterocyclic ring(s).
- (1) Adenine
 - (2) Thymine
 - (3) Cholesterol
 - (4) Guanine
50. **Assertion (A):** Relaxation of the diaphragm and external inter-costal muscles reduces the pulmonary volume.
Reason (R): Return of diaphragm and sternum back to their original positions leads to a decrease in intrapulmonary pressure to slightly below the atmospheric pressure.
 In light of above statements, select the correct option.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true but (R) is false.
 - (4) Both (A) and (R) are false.
51. Select the odd one w.r.t. organised endocrine glands present in human body.
- (1) Pancreas
 - (2) Testis
 - (3) Thymus
 - (4) Kidney
52. Consider the given structures
- a. Testis
 - b. Adrenal gland
 - c. Hypothalamus
 - d. Ovary
- Select the option representing paired structures which secrete atleast one steroidal hormone.
- (1) a and d only
 - (2) a, b and d
 - (3) a, c and d
 - (4) b and c only
53. 'X' and 'Y' are the lymphoid organs where immature lymphocytes differentiate into antigen-sensitive lymphocytes. 'Y' is present in long bones assisting in erythropoiesis while 'X' is a lobed organ. Which of the following options is **incorrect** w.r.t 'X'?
- (1) T-lymphocytes migrate to organs like spleen, lymph nodes, tonsils, etc, after maturation in 'X'.
 - (2) The size of 'X' reduces in older individuals due to its degeneration.
 - (3) 'X' is a paired secondary lymphoid organ
 - (4) 'X' is located behind breastbone on the ventral side of aorta.
54. In a monomeric antibody molecule, antigen binding sites are present on
- (1) C-terminal of both light and heavy chains
 - (2) N-terminal of both light and heavy chains
 - (3) N-terminal of light chain and C-terminal of heavy chain
 - (4) C-terminal of light chain and N-terminal of heavy chain

55. Select the **incorrect** option w.r.t. immunity in humans.

- (1) Innate immunity – Non-specific defence mechanism present since birth
- (2) Acquired immunity – Highly specific defence mechanism that develops after exposure to pathogen
- (3) Active immunity – Short-lived and provides immediate immune response
- (4) Passive immunity – Protection against pathogens using readymade antibodies

56. Complete the analogy by selecting the **correct** option.

Acromegaly : Growth hormone :: Graves' disease : _____.

- (1) Growth hormone
- (2) Vasopressin
- (3) Tetraiodothyronine
- (4) Insulin

57. Which of the following features is common among *Catla*, *Trygon*, *Salamandra* and *Myxine*?

- (1) Presence of air bladder
- (2) Absence of scales
- (3) Marine habitat
- (4) Presence of notochord at some stage(s) of their life

58. If we compare the volume of the filtrate formed per day with that of the urine released per day in a healthy adult human under normal physiological conditions, then the ratio obtained will be most likely

- (1) 99 : 1
- (2) 100 : 1
- (3) 120 : 1
- (4) 200 : 1

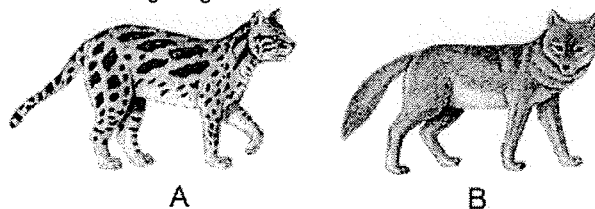
59. Which of the following was the biggest land reptile and had huge fearsome dagger-like teeth?

- (1) *Pteranodon*
- (2) *Tyrannosaurus*
- (3) *Triceratops*
- (4) *Stegosaurus*

60. 'The evolution of life forms had occurred but driven by use and disuse of organs'. This statement is associated with

- (1) Darwin
- (2) Hugo deVries
- (3) Wallace
- (4) Lamarck

61. Observe the figure given below



Choose the **correct** answer from the options given below.

- (1) 'A' is an Australian marsupial that exhibits convergent evolution with Tasmanian wolf
- (2) 'A' is a placental mammal that exhibits convergent evolution with 'B'
- (3) 'B' is a placental mammal that exhibits convergent evolution with 'A'
- (4) Both 'A' and 'B' are placental mammals that exhibit divergent evolution with each other

62. Under normal physiological conditions, distal convoluted tubule of nephron performs conditional reabsorption of Na^+ and water. It also reabsorbs

- (1) NH_3
- (2) Uric acid
- (3) H^+
- (4) HCO_3^-

63. *Limulus* is an aquatic organism that has remained relatively unchanged over the time and respire through

- (1) Flame cells
- (2) Malpighian tubules
- (3) Green glands
- (4) Book gills

64. If a researcher in a lab isolated the protoplasts of both onion and garlic cells and fused them to form hybrids for obtaining characteristics of both, then this process is called

- (1) Genetic cloning
- (2) Somatic hybridisation
- (3) Micropropagation
- (4) Bioprocess engineering

65. The vector used for transfer of nematode specific genes in tobacco plants to mediate the process of RNA interference is disarmed

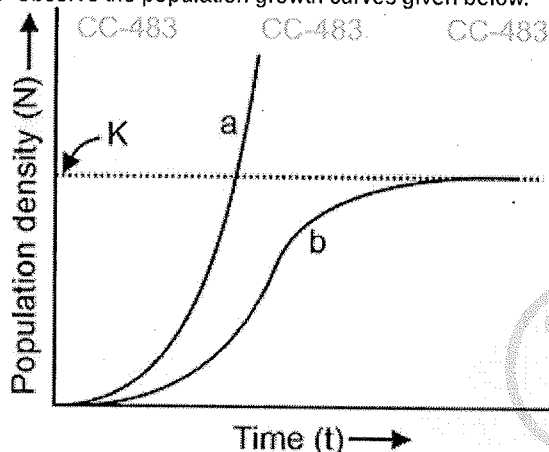
- (1) *Meloidogyne incognita*
- (2) *Agrobacterium tumefaciens*
- (3) *E. coli*
- (4) *Bacillus thuringiensis*

66. Match the following columns and select the correct option.

Column - I	Column - II
(a) Camembert cheese	(i) <i>Leuconostoc</i>
(b) Swiss Cheese	(ii) <i>Penicillium</i>
(c) Dosa	(iii) <i>Propionibacterium</i>
(d) Pectinase	(iv) <i>Aspergillus</i>

- (1) (a-ii), (b-iii), (c-iv), (d-i)
- (2) (a-ii), (b-iii), (c-i), (d-iv)
- (3) (a-ii), (b-i), (c-iii), (d-iv)
- (4) (a-iv), (b-iii), (c-i), (d-ii)

67. Observe the population growth curves given below.



Choose the correct equations for 'a' and 'b' respectively.

- (1) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$
- (2) $N_t = N_0 e^{rt}$; $\frac{dN}{dt} = rN$
- (3) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$; $\frac{dN}{dt} = rN$
- (4) $\frac{dN}{dt} = rN$; $N_t = N_0 e^{rt}$

68. Plants with Kranz anatomy

- (1) Lack photochemical phase of photosynthesis
- (2) Show lower temperature optimum for photosynthesis as compared to the plants without Kranz anatomy
- (3) Show oxygenase activity of RuBisCO in bundle sheath cell
- (4) Utilise the same amount of NADPH as bell pepper for each molecule of CO_2 fixed

69. All of the following statements are true for non-cyclic photophosphorylation, except

- (1) It involves synthesis of both ATP and NADPH
- (2) It mostly occurs in stroma lamellae membrane
- (3) It requires an external electron donor
- (4) It is connected with photolysis of water

70. In the TCA cycle, the first decarboxylation reaction occurs during the conversion of

- (1) Succinyl CoA to succinic acid
- (2) Oxaloacetic acid to citric acid
- (3) Oxalosuccinic acid to α -ketoglutaric acid
- (4) Succinic acid to malic acid

71. In an ecosystem, 200 kJ of sunlight energy falls on the organism that occupies the first trophic level of a food chain. What amount of this energy will be utilised by the organism for its different biological activities which occupies the third trophic level in this food chain?

- (1) 2 J

(2) 1.8 J

- (3) 20 J

- (4) 18 J

72. During alcoholic fermentation,

- (1) Ethyl alcohol is converted into carbon dioxide and water
- (2) NAD^+ is reoxidised to $\text{NADH} + \text{H}^+$
- (3) All the energy of glucose is released
- (4) Acetaldehyde is reduced to alcohol by $\text{NADH} + \text{H}^+$

73. The rate of decomposition of detritus is faster in the ecosystem due to the following factors, except

- (1) Presence of aerobic soil microbes
- (2) Detritus rich in lignin and chitin
- (3) Warm and moist environment
- (4) Detritus rich in sugar

74. Select the incorrect statement.

- (1) Families are characterised on the basis of both vegetative and reproductive features of plant species.
- (2) Species comprises a group of related genera which has more characters in common in comparison to genera of other species
- (3) Convolvulaceae and Solanaceae are included in the order polmoniales
- (4) Classes with a few similar characters are assigned to a higher category called division

75. Which of the following fungi is used extensively in biochemical and genetic work?

- (1) *Aspergillus*
- (2) *Claviceps*
- (3) *Neurospora*
- (4) *Agaricus*

76. In which of the following flowers, margins of petals overlap one another but not in any particular direction?

- (1) *Cassia*
- (2) *Calotropis*
- (3) Pea
- (4) Cotton

77. Select the **incorrect** statement w.r.t. members of family Liliaceae, such as *Allium*.

- (1) Fruit is capsule, rarely berry
- (2) Flower is unisexual and zygomorphic
- (3) Seed is endospermous
- (4) Ovary is trilocular with many ovules

78. Read the following statements (A-D).

- (A) In *Dianthus* and *Primrose*, ovules are borne on central axis and septa are absent.
- (B) Variation in the length of filaments of stamens in flower is seen in *Salvia* and mustard.
- (C) In sunflower and marigold, the placenta forms a ridge along the ventral suture of the ovary and ovules are borne on this ridge.
- (D) In mustard, ovary is one-chambered but it becomes two chambered due to the formation of false septum.

The **correct** statements are

- (1) Only (A) and (D)
- (2) Only (B), (C) and (D)
- (3) Only (A), (B) and (D)
- (4) (A) and (C)

79. State true (T) or false (F) for the given statements and select the **correct** option.

- (A) Fungi are heterotrophic and absorb soluble organic matter from dead substrate.
- (B) Bladderwort and Venus fly trap are insectivorous plants and *Cuscuta* is a parasite.
- (C) In *Agaricus* and *Puccinia*, mycelium is generally aseptate and coenocytic.
- (D) Fusion of protoplasm between two motile or non-motile gametes is called plasmogamy.

- (1) (A)T, (B)T, (C)F, (D)T
- (2) (A)F, (B)F, (C)T, (D)F
- (3) (A)T, (B)F, (C)T, (D)F
- (4) (A)F, (B)T, (C)F, (D)T

80. Read the following statements and select the **correct** option.

Statement A: The viruses are non-cellular and are characterised by having an inert crystalline structure outside the living cell.

Statement B: A virus is a nucleoprotein and its genetic material is infectious.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both the statements A and B are correct
- (4) Both the statements A and B are incorrect

81. **Assertion (A):** ELISA is based on the principle of antigen-antibody interaction.

Reason (R): Infection by pathogen can be detected only by detecting the antibodies synthesised against the pathogen. In the light of above statements, choose the **correct** option

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) Both (A) and (R) are false
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

82. Select the **incorrect** match.

- (1) Cuttle fish - Presence of a file-like rasping organ in the mouth
- (2) Prawn - Presence of a balancing organ
- (3) Scorpion - Possesses book lungs for respiration
- (4) Roundworm - Possesses both male and female sex organs in a single body

83. In male frogs, testes are found adhered to the upper part of kidneys by a double fold of peritoneum called

- (1) Mesovarium
- (2) Mesorchium
- (3) Endometrium
- (4) Myometrium

84. In humans, the region present in between seminiferous tubules of testes contain all of the following cells/structures, **except**

- (1) Small blood vessels
- (2) Interstitial cells
- (3) Immunologically competent cells
- (4) Sertoli cells

85. In human females, formation of first polar body occurs in

- (1) Fallopian tubes
- (2) Uterus
- (3) Cervix
- (4) Ovary

86. Which among the following is an unpaired sex accessory gland in human males?

- (1) Epididymis
- (2) Seminal vesicle
- (3) Prostate gland
- (4) Bulbourethral gland

87. In male cockroaches, genital pouch lies at the A of the abdomen and is bounded dorsally by B terga and ventrally by the C sternum.

Identify A, B and C and select the option which fills the blanks correctly.

	A	B	C
(1)	Proximal end	7 th and 8 th	9 th
(2)	Anterior end	9 th and 10 th	10 th
(3)	Hind end	9 th and 10 th	9 th
(4)	Posterior end	7 th , 8 th and 9 th	7 th

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

88. Read the following statements carefully w.r.t. human sperm.

- (a) It is composed of head, neck, middle piece and a tail.
- (b) The anterior portion of acrosome is covered by a cap-like nucleus.
- (c) A plasma membrane envelops whole body of sperm except the head region.
- (d) Spirally arranged mitochondria are present in the middle piece.

Choose the option with only correct statement(s).

- (1) (a) and (b)
- (2) (b) and (c)
- (3) (c) and (d)
- (4) (a) and (d)

89. Listed below are functions/effects of certain contraceptives.

- a. Inhibits ovulation
 - b. Makes endometrium unsuitable for implantation
 - c. Alters the quality of cervical mucus
- All the above mentioned functions/effects are true for

- (1) Diaphragm
- (2) Steroidal oral contraceptive pills
- (3) Multiload 375
- (4) Non-medicated IUD

90. Match column I with column II.

(a)	LNG-20	(i)	Hormone releasing IUD
(b)	Nirodh	(ii)	Supresses the motility and fertilising capacity of the sperms
(c)	Cu7	(iii)	Barrier method of contraception
(d)	Tubectomy	(iv)	Highly effective contraceptive method but very poor reversibility

Select the correct option.

- (1) a(i), b(iii), c(ii), d(iv)
- (2) a(i), b(iii), c(iv), d(ii)
- (3) a(iv), b(ii), c(iii), d(i)
- (4) a(iii), b(i), c(iv), d(ii)

91. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

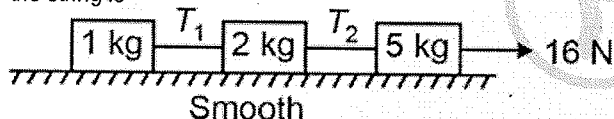
Assertion (A): When a neutral body is positively charged, its mass decreases.

Reason (R): A body acquires positive charge when it loses electrons.

In the light of the above statements, the correct option is

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is true but (R) is false
 - (4) Both (A) and (R) are false
92. A coil has a self-inductance of 0.02 H. The current through it, is allowed to change at the rate of 4 A in 2×10^{-2} s. The e.m.f induced in the coil will be
- (1) 4 V
 - (2) 2 V
 - (3) 6 V
 - (4) Zero

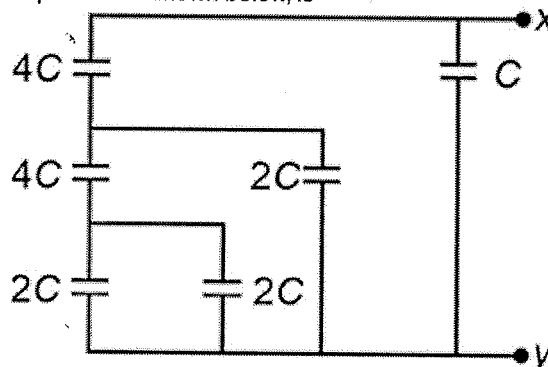
93. Three masses 1 kg, 2 kg and 5 kg kept on a smooth horizontal table, are connected by light, inextensible strings and are pulled by a 16 N force as shown. The tension T_1 in the string is



- (1) 10 N
 - (2) 24 N
 - (3) 2 N
 - (4) 6 N
94. Four electric charges of $8 \mu\text{C}$, $5 \mu\text{C}$, $-3 \mu\text{C}$, $-10 \mu\text{C}$ are placed at the corners of a square of side $\sqrt{2}$ m. The potential at the centre of the square is

- (1) $9 \times 10^3 \text{ V}$
- (2) Zero
- (3) $1.8 \times 10^5 \text{ V}$
- (4) $2.7 \times 10^3 \text{ V}$

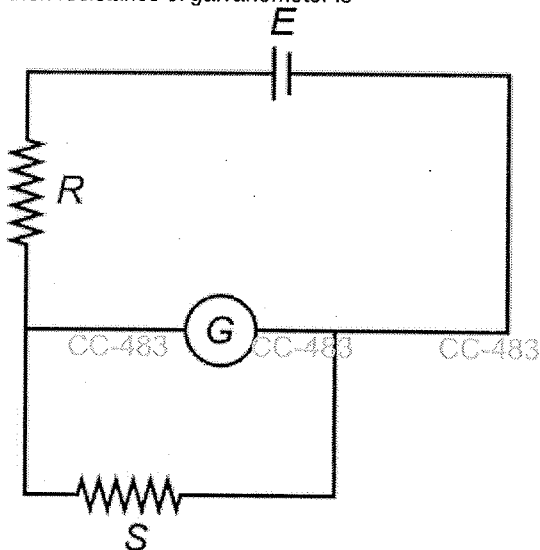
95. The equivalent capacitance of the combination of capacitors as shown below, is



- (1) $7C$
 - (2) $9C$
 - (3) $\frac{C}{2}$
 - (4) $3C$
96. The drift velocity of free electrons in a conductor is v when a current i is flowing in it. If both the area of cross-section and current are doubled, then drift velocity will be

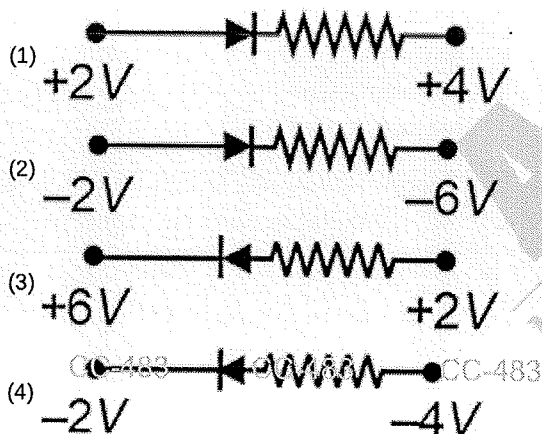
- (1) $2v$
- (2) $\frac{v}{2}$
- (3) v
- (4) $4v$

97. In an experiment to determine the resistance of a galvanometer by half deflection method, the following circuit is used. In one set of reading, if $R = 1000 \Omega$ and $S = 20 \Omega$ then resistance of galvanometer is



- (1) $\frac{1000}{49} \Omega$
 (2) $\frac{1000}{9} \Omega$
 (3) $\frac{250}{11} \Omega$
 (4) $\frac{1000}{11} \Omega$

98. A forward biased diode among the following is



99. A silicon (Si) specimen is doped with aluminium (Al). The concentration of acceptor atoms is 10^{18} m^{-3} . Given that the intrinsic carrier concentration is 10^{16} m^{-3} , the concentration of electrons in the specimen is

- (1) 10^{14} m^{-3}
 (2) 10^{16} m^{-3}
 (3) 10^{18} m^{-3}
 (4) 10^{12} m^{-3}

100. The potential difference that must be applied to stop the fastest moving photoelectrons emitted by a metal surface, having work function 4.5 eV, when ultraviolet light of 2000 Å falls on it, will be

- (1) -0.7 V
 (2) -1.7 V
 (3) -1.2 V
 (4) -0.8 V

101. An electron in the hydrogen atom jumps from state n to the ground state. The wavelength so emitted illuminates a photo-sensitive material having work function 4.09 eV. If the stopping potential of the photoelectrons is 8 V, the value of n is

- (1) 5
 (2) 2
 (3) 3
 (4) 4

102. When $^{235}_{92}\text{U}$ undergoes fission, 0.2% of its original mass is changed into energy. How much energy is released if 1 kg of $^{235}_{92}\text{U}$ undergoes fission?

- (1) $18 \times 10^{11} \text{ J}$
 (2) $6 \times 10^5 \text{ J}$
 (3) $18 \times 10^{13} \text{ J}$
 (4) $18 \times 10^{10} \text{ J}$

103. A body of mass 1 kg travels in a straight line with velocity $v = \alpha x^{3/2}$ where $\alpha = 10 \text{ m}^{-1/2} \text{ s}^{-1}$. The work done by the net force during its displacement from $x = 1 \text{ m}$ to $x = 3 \text{ m}$ is

- (1) 50 J
 (2) 100 J
 (3) 1300 J
 (4) 500 J

104. A body of mass 10 kg moves along a smooth inclined plane ($\theta = 30^\circ$) with constant velocity through a distance of 10 m. Calculate the work done by applied external force to move the body through desired distance of 10 m.

- (1) 100 J
 (2) 150 J
 (3) 500 J
 (4) 400 J

105. Consider the following statements:

- (A) The electromotive force (EMF) of a cell is equal to the potential difference across its terminals when no current is flowing.
 (B) Voltage drop happens in a cell due to internal resistance when current flows through it.
 (C) If multiple identical cells are connected in series, the total EMF decreases.
 (D) If multiple identical cells are connected in parallel with their positive terminals on the same side, the total EMF remains same as individual cell.

Which of the above statement(s) is/are correct?

- (1) Both (A) and (C)
 (2) Both (B) and (C)
 (3) (A), (B) and (D)
 (4) (B), (C) and (D)

106. An object of mass 2 kg is placed on a smooth horizontal surface. A water jet throws water at a rate of 2 kg/s with speed 20 m/s which strikes the object horizontally. The force applied by water jet and acceleration of the object respectively, are

- (1) 20 N, 40 m/s²
 (2) 40 N, 20 m/s²
 (3) Zero, zero
 (4) 12 N, 12 m/s²

107. A vehicle travels half of the total distance with speed 2 m/s and the other half with speed 5 m/s, then its average speed is

- (1) $\frac{7}{2}$ m/s
 (2) $\frac{20}{7}$ m/s
 (3) $\frac{14}{3}$ m/s
 (4) $\frac{7}{20}$ m/s

108. A particle moving with uniform acceleration crosses two points A and B present in a straight line with speed 10 m/s and 20 m/s respectively, the speed of particle at mid-point of A and B will be

- (1) $5\sqrt{10}$ m/s
 (2) $10\sqrt{5}$ m/s
 (3) $\sqrt{5}$ m/s
 (4) $\sqrt{10}$ m/s

109. In the given expression of force $F = a \log\left(\frac{b}{x}\right)$ where x is displacement, the dimensions of a and b respectively are

- (1) $[a] = [MLT^{-2}]$, $[b] = [L]$
 (2) $[a] = [M^{-1}]$, $[b] = [L]$
 (3) $[a] = [ML^{-1}]$, $[b] = [MLT^{-1}]$
 (4) $[a] = [MT^{-2}]$, $[b] = [ML^2]$

110. The side of a cube is (2.00 ± 0.01) cm. The volume and total surface area of cube respectively are

- (1) (8.00 ± 0.12) cm³, (24.0 ± 0.24) cm²
 (2) (8.00 ± 0.01) cm³, (24.0 ± 0.01) cm²
 (3) (8.00 ± 0.04) cm³, (24.0 ± 0.06) cm²
 (4) (8.00 ± 0.03) cm³, (24 ± 0.02) cm²

111. A body oscillates in SHM according to the equation $x = 10 \cos\left(2\pi t + \frac{\pi}{4}\right)$ cm. Its instantaneous displacement at $t = 1$ s is

- (1) 10 cm
 (2) $\frac{5}{\sqrt{2}}$ cm
 (3) $5\sqrt{2}$ cm
 (4) $10\sqrt{2}$ cm

112. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The presence of moisture increases the velocity of sound in air.

Reason (R): Density of moist air is more than the density of dry air.

In the light of the above statements, the correct option is

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false

113. In a diffraction pattern due to a single slit of width a , the first minimum is observed at an angle 30° when light of wavelength λ is incident on the slit. The first secondary maximum is observed at an angle.

- (1) $\sin^{-1}\left(\frac{1}{2}\right)$
 (2) $\sin^{-1}\left(\frac{3}{4}\right)$
 (3) $\sin^{-1}\left(\frac{1}{4}\right)$
 (4) $\sin^{-1}\left(\frac{2}{3}\right)$

114. Match List-I with List-II.

	List-I (Types of EM waves)		List-II (Application)
a.	Infrared rays	(i)	Water purifier
b.	Microwaves	(ii)	Remote switches
c.	UV rays	(iii)	Used in medicine to destroy cancer cells
d.	Gamma rays	(iv)	Cooking

(1) a(ii), b(iii), c(iv), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(i), c(ii), d(iv)

 115. A steel rod has a radius of 10 mm and a length of 1 m. A 100 kN force stretches it along its length. The elongation in rod is (Assume young's modulus $Y = 2 \times 10^{11} \text{ N/m}^2$)

(1) 0.8 mm

(2) 1.59 mm

(3) 3.8 mm

(4) 2.1 mm

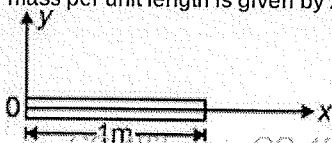
 116. Two objects A & B are moving in a plane with velocities $\vec{v}_A = (3\hat{i} + 4\hat{j}) \text{ m/s}$ and $\vec{v}_B = (7\hat{i} - 3\hat{j}) \text{ m/s}$ respectively, the velocity of object A with respect to object B, will be (in m/s)

 (1) $4\hat{i} - 7\hat{j}$

 (2) $4\hat{i} + \hat{j}$

 (3) $7\hat{i} + 7\hat{j}$

 (4) $-4\hat{i} + 7\hat{j}$

 117. The centre of mass of the straight rod given below if its mass per unit length is given by $\lambda = (3x) \text{ kg m}^{-1}$, will be at

 (1) $x = \frac{1}{2} \text{ m}$

 (2) $x = \frac{2}{3} \text{ m}$

 (3) $x = \frac{3}{4} \text{ m}$

 (4) $x = 1 \text{ m}$

118. Consider the following statements A and B, identify the correct answer:

A: In a perfectly rigid body, the net positive work done by external torques increase the rotational kinetic energy of the body

B: Angular acceleration of a rotating body having fixed axis of rotation is inversely proportional to the moment of inertia of the body for a given torque.

(1) A is correct but B is incorrect

(2) A is incorrect but B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

 119. Two masses each equal to M are moving on a circular path of radius R about another fixed mass M (at centre of circular path). The gravitational potential energy of the system is

 (1) $-\frac{GM^2}{2R}$

 (2) $-\frac{GM^2}{R}$

 (3) $-\frac{2GM^2}{R}$

 (4) $-\frac{5GM^2}{2R}$

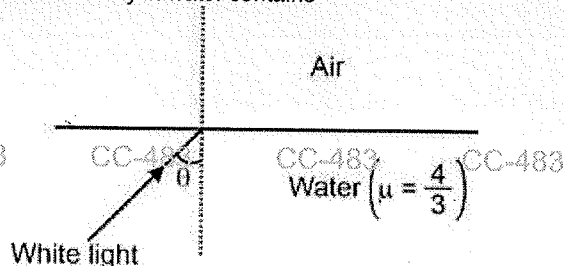
120. An object is weighed at the equator using a physical balance and a spring balance. When the same object is taken to the pole, then corresponding readings on the physical balance and spring balance (also taken there) will respectively

(1) Remain same, increase

(2) Increase, remain same

(3) Both remain same

(4) Both increase

 121. A beam of white light, falls on the interface of water-air as shown. If critical angle for green light is $\theta = \sin^{-1}\left(\frac{3}{4}\right)$, then reflected ray in water contains


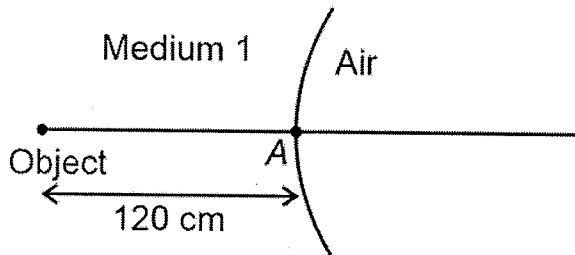
(1) Violet, Indigo, Blue only

(2) Yellow, Orange, Red only

(3) No colour

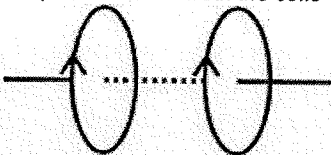
(4) White light

122. A circular arc of radius 40 cm is separating air and medium 1 of refractive index of 1.5 as shown. The final image will form, as measured from point A, at



- (1) 40 cm
(2) 50 cm
(3) 120 cm
(4) Infinity
123. Consider the following statements:
(A) In a region of uniform electric field, the net charge contained in a volume is zero.
(B) Gauss law gives incorrect result for uniform electric field.
The correct option(s) is/are
- (1) Only (A)
(2) Only (B)
(3) Both (A) and (B)
(4) Neither (A) nor (B)

124. The current is flowing in two co-axial coils in same direction as shown in figure. On decreasing the distance between the two, the current in both the coils



- (1) Increases
(2) Decreases
(3) Remains same
(4) Information incomplete
125. A uniform bar magnet has magnetic moment M . If it is cut into 4 equal parts parallel to its length such that width of each part is same, then magnetic moment of each part will be

- (1) $\frac{M}{4}$
(2) $\frac{M}{3}$
(3) $\frac{M}{5}$
(4) M

126. A circular coil of radius R having current I is placed in uniform magnetic field B . If angle between area vector of coil and magnetic field is 60° , then torque on the coil will be

- (1) $\frac{\pi R^2 IB}{2}$
(2) $\frac{\sqrt{3}\pi R^2 IB}{2}$
(3) $\pi R^2 IB$
(4) Zero

127. If a proton has velocity $(2\hat{i} + 3\hat{k})$ m/s and it is subjected to magnetic field of $4\hat{i}$ T, then its

- (1) Speed will not change
(2) Path will not change
(3) Velocity will remain same
(4) Momentum will remain same

128. The value of capacitance C in a LCR series circuit with inductance $L = 50$ mH, connected with voltage source of angular frequency 100 rad/s so that current in circuit becomes maximum, is

- (1) 1 mF
(2) 2 mF
(3) 1 μ F
(4) 4 μ F

129. In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{3}$. The ratio of $I_s : I_p$ is equal to (symbols carry their usual meaning)

- (1) 1 : 3
(2) 1 : 1
(3) 9 : 1
(4) 2 : 1

130. Consider the following statements:

Statement A: When a capillary tube is dipped into a liquid, if the liquid neither rises or falls in the capillary, if then the angle of contact must be zero.

Statement B: The working of a Venturimeter is based on the Bernoulli's theorem.

Based upon above information pick correct option.

- (1) Both statement (A) and (B) are correct
(2) Both statement (A) and (B) are incorrect
(3) Statement (A) is correct while statement (B) is incorrect
(4) Statement (A) is incorrect while statement (B) is correct

131. A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0 . After sometime, the ball falls with constant velocity. The net force acting on the ball after it attains constant velocity, will be

- (1) $mg\left(\frac{\rho_0}{\rho}-1\right)$
- (2) $mg(\rho-\rho_0)$
- (3) $mg\left(1-\frac{\rho}{\rho_0}\right)$
- (4) Zero

132. The temperature of 100 g of water is to be raised from 30 °C to 90 °C by adding steam to it. The mass of steam required to raise this temperature will be nearly (Take, $S_w = 1 \text{ cal/g } ^\circ\text{C}$, $L = 540 \text{ cal/g}$)

- (1) 11 g
- (2) 15 g
- (3) 18 g
- (4) 7.5 g

133. The internal energy of an ideal monoatomic gas increases by the same amount as work done on the gas, then

- (1) The process should be adiabatic
- (2) The process should be isothermal
- (3) The process should be isochoric
- (4) The process should be isobaric

134. The work done by 3 moles of gas at 47 °C to triple its volume at constant pressure is ($R = 2 \text{ cal mol}^{-1} ^\circ\text{C}^{-1}$)

- (1) 3402 cal
- (2) 3428 cal
- (3) 3832 cal
- (4) 3840 cal

135. The ratio of total K.E of a molecule of Argon and Oxygen gas at 27 °C is equal to

- (1) 3 : 5
- (2) 3 : 2
- (3) 2 : 3
- (4) 5 : 7

CHEMISTRY

136. Consider the following statements

- (a) The solutions which show large negative deviation from Raoult's law form minimum boiling azeotrope.
- (b) Two solutions having same osmotic pressure at a given temperature are called isotonic solution.
- (c) K_f is independent of solute concentration.

The correct statement(s) are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

137. Match List-I (Salts with degree of dissociation α) with List-II (van't Hoff factor (i)).

- | List-I | List-II |
|---|-----------|
| a. NaCl ($\alpha = 90\%$) | (i) 3.25 |
| b. CaCl_2 ($\alpha = 60\%$) | (ii) 2.2 |
| c. AlCl_3 ($\alpha = 50\%$) | (iii) 1.9 |
| d. $\text{K}_3[\text{Fe}(\text{CN})_6]$ ($\alpha = 75\%$) | (iv) 2.5 |

The correct option is

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iii), b(ii), c(iv), d(i)
- (3) a(ii), b(i), c(iii), d(iv)
- (4) a(iv), b(iii), c(ii), d(i)

138. Given below are two statements:

Statement I: The greater the charge on the cation, the greater the covalent character of the ionic bond.

Statement II: The smaller the size of the cation and larger the size of the anion, the greater the covalent character of an ionic bond

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statement I and statement II are incorrect

139. Number of planar species among the following are

I_3^- , ClF_3 , BF_3 , XeO_3 , B_2H_6

- (1) 2
- (2) 3
- (3) 4
- (4) 5

140. The correct order of bond order is

- (1) $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$
- (2) $\text{O}_2^{2-} < \text{O}_2^- < \text{O}_2 < \text{O}_2^+$
- (3) $\text{O}_2^{2-} < \text{O}_2 < \text{O}_2^- < \text{O}_2^+$
- (4) $\text{O}_2 < \text{O}_2^+ < \text{O}_2^{2-} < \text{O}_2^-$

141. The correct *d*-orbital configuration in octahedral complex $[\text{Co}(\text{CN})_6]^{3-}$ is

- (1) $t_{2g}^4 e_g^2$
- (2) $t_{2g}^6 e_g^0$
- (3) $t_{2g}^2 e_g^4$
- (4) $t_{2g}^3 e_g^3$

142. Given below are two statements:

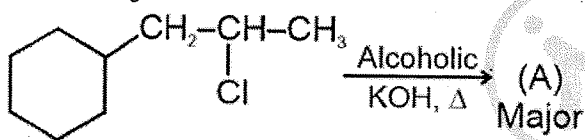
Statement I: $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]^{3+}$ complex shows geometrical isomerism.

Statement II: $[\text{Co}(\text{en})_3]^{3+}$ complex shows optical isomerism.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are true
- (2) Both statement I and statement II are false
- (3) Statement I is true but statement II is false
- (4) Statement I is false but statement II is true

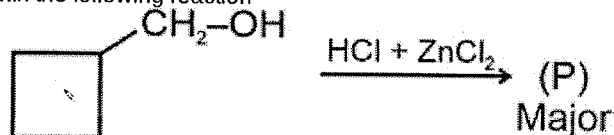
143. In the following reaction



The product (A) will be

- (1)
- (2)
- (3)
- (4)

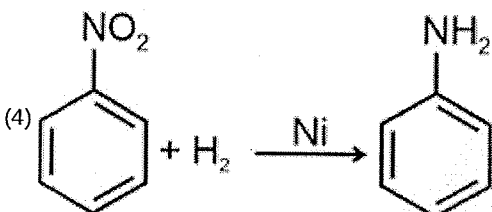
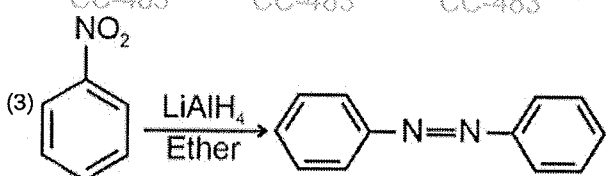
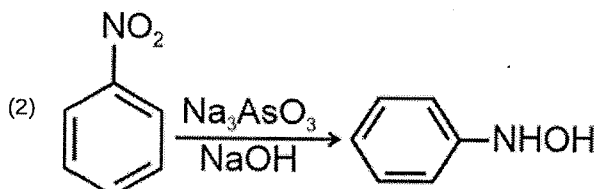
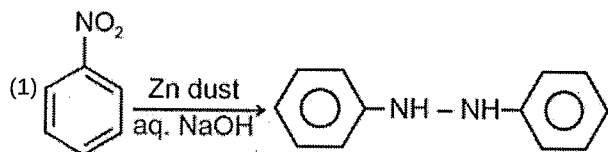
144. In the following reaction



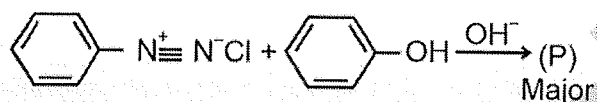
The product (P) will be

- (1)
- (2)
- (3)
- (4)

145. Choose the incorrect match for the major product in given reactions



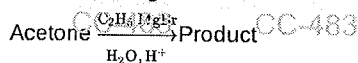
146. In the following reaction



The colour of product (P) will be

- (1) Yellow
- (2) Green
- (3) Orange
- (4) Blue

147. What is the IUPAC name of the organic compound formed in the following chemical reaction?

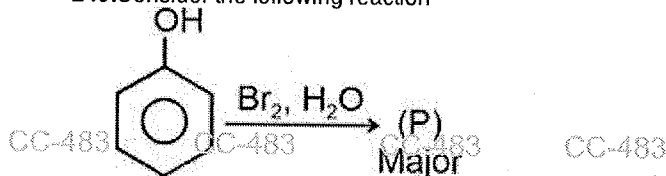


- (1) 2-Methylbutan-2-ol
- (2) 2-Methylpropan-2-ol
- (3) Pentane-2-ol
- (4) Pentane-3-ol

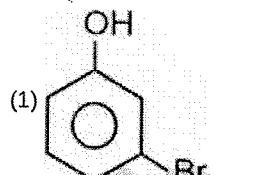
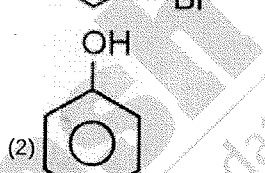
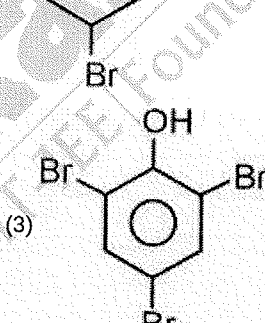
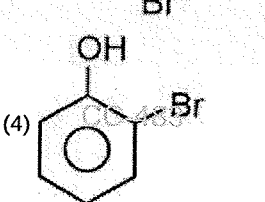
148. If the given molecules are moving with the same velocity than which will be associated with longest de Broglie wavelength?

- (1) H_2
- (2) N_2
- (3) O_2
- (4) CO_2

149. Consider the following reaction



The product 'P' is

- (1) 
- (2) 
- (3) 
- (4) 

150. Which of the following reagents can not be used to oxidise primary alcohol to aldehyde only?

- (1) PCC
- (2) Anhydrous CrO_3
- (3) Heat in the presence of Cu at 573 K
- (4) KMnO_4 in acidic medium

151. Match List-I with List-II.

	List-I (Chemical reaction)		List-II (Name of process)
a.	$4\text{NH}_3 + 5\text{O}_2 \xrightarrow[\text{Rh}]{500\text{ K, 9 bar}} 4\text{NO} + 6\text{H}_2\text{O}$	(i)	Contact process
b.	$4\text{HCl} + \text{O}_2 \xrightarrow[\text{CuCl}_2]{723\text{ K}} 2\text{Cl}_2 + 2\text{H}_2\text{O}$	(ii)	Ostwald process
c.	$\text{N}_2 + 3\text{H}_2 \xrightarrow{\text{Fe+Mo}} 2\text{NH}_3$	(iii)	Deacons' process
d.	$2\text{SO}_2 + \text{O}_2 \xrightarrow[\text{V}_2\text{O}_5]{673-723\text{ K}} 2\text{SO}_3$	(iv)	Haber's process

Choose the correct answer from the options given below.

- (1) a(ii), b(iii), c(iv), d(i)
- (2) a(i), b(iii), c(iv), d(ii)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(ii), b(iii), c(i), d(iv)

 152. The number of P—OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($\text{H}_4\text{P}_2\text{O}_7$) respectively are:

- (1) Five and four
- (2) Four and four
- (3) Four and five
- (4) Five and five

153. The compound of Xenon with zero dipole moment is

- (1) XeO_3
- (2) XeF_4
- (3) XeOF_4
- (4) XeO_2F_2

154. Select the correct statement among the following

- (1) The lanthanoid ions, La^{3+} and Lu^{3+} are paramagnetic.
- (2) Pr, Nd and Tb exhibit +4 oxidation state but only in oxides
- (3) Thorium only exhibits +3 and +4 oxidation state
- (4) In Wacker process, the oxidation of ethyne to ethanal is catalysed by $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$

155. Given below are two statements:

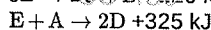
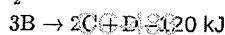
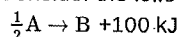
Statement I: Dichromate ion in acidic medium gets converted to chromate ions.

Statement II: Acidified permanganate solution oxidises iodides to free iodine.

In the light of the above statements, choose the most appropriate answer from the options given below:

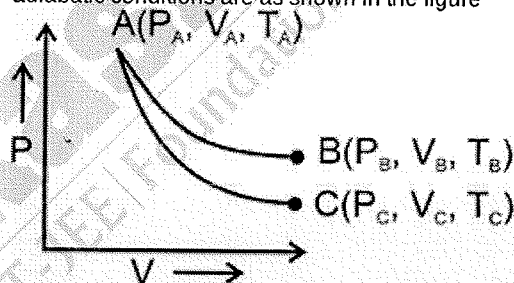
- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statement I and statement II are incorrect

156. Consider the following process


 For, $\text{B} + \text{D} \rightarrow \text{E} + 2\text{C}$; ΔH will be

- (1) 305 kJ mol^{-1}
- (2) 245 kJ mol^{-1}
- (3) -305 kJ mol^{-1}
- (4) -245 kJ mol^{-1}

157. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure



Identify the correct option regarding the above-mentioned P - V curve.

- (1) AB is an adiabatic expansion process
- (2) $\Delta S_{AB} > \Delta S_{AC}$
- (3) AC an isothermal expansion process
- (4) $T_C = T_B$

158. Number of molecules of water is maximum in

- (1) 36 mL of water
- (2) 0.36 g of water
- (3) 10^{-3} mol of water
- (4) 0.00224 L of water vapours at 1 atm and 273 K

159. Match List-I with List-II.

	List-I (Atomic number)		List-II (Symbol)
a.	102	(i)	Uuq
b.	115	(ii)	Uns
c.	114	(iii)	Unb
d.	107	(iv)	Uup

Choose the correct answer from the options given below.

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(iv), b(iii), c(i), d(ii)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(ii), b(i), c(iii), d(iv)

 160. If standard reduction potential of A^{2+}/A , B^{2+}/B and C^{2+}/C are 0.52 V, -0.14 V and -2.93 V respectively, then reducing power of the metals will be in order of

- (1) $A > C > B$
- (2) $A > B > C$
- (3) $B > A > C$
- (4) $C > B > A$

 161. On hydrolysis of sucrose, equilibrium constant K_C of the reaction is 2×10^{13} at 300 K. ΔG° at 300K will be

- (1) $2.16 \times 10^6 \text{ J mol}^{-1}$
- (2) $-7.64 \times 10^4 \text{ J mol}^{-1}$
- (3) $-2.67 \times 10^4 \text{ J mol}^{-1}$
- (4) $-1.26 \times 10^3 \text{ J mol}^{-1}$

162. Which among the following ions does not have noble gas configuration?

- (1) F^-
- (2) Zn^{2+}
- (3) Na^+
- (4) Ca^{2+}

 163. If molar conductance of 0.02 M solution of weak monobasic acid is $10 \text{ S cm}^2 \text{ mol}^{-1}$ and at infinite dilution is $250 \text{ S cm}^2 \text{ mol}^{-1}$ then the dissociation constant of the acid will be

- (1) 1.6×10^{-8}
- (2) 3.2×10^{-5}
- (3) 3.2×10^{-6}
- (4) 1.6×10^{-2}

164. Given below are two statements:

Statement I: 50 mL, 0.1 M HCl and 100 mL, 0.1 M CH_3COONa form acidic buffer solution.

Statement II: In the following equilibrium, $\text{S}_8(\text{s}) + 8\text{O}_2(\text{g}) \rightleftharpoons 8\text{SO}_2$ at temperature T, $K_p = K_C$.

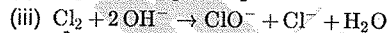
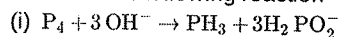
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statement I and statement II are incorrect

165. Choose the incorrect order of group 14 elements with respect to the given properties

- (1) $\text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$ (first ionisation enthalpy)
- (2) $\text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$ (Boiling point)
- (3) $\text{Si} > \text{Ge} > \text{Pb} > \text{Sn}$ (Melting point)
- (4) $\text{Pb} > \text{Si} = \text{Ge} = \text{Sn}$ (electronegativity)

166. Consider the following reaction



The disproportionation reaction is/are

- (1) (i) and (ii) only
- (2) (iii) only
- (3) (ii) and (iii) only
- (4) (i), (ii) and (iii)

167. Consider the following statements

- (a) $\text{C}_3\text{H}_6\text{O}$ does not show metamerism
- (b) Ethyl carbocation is less stable as compared to methyl carbocation.
- (c) Hyperconjugation is a temporary effect
- (d) All the carbon atoms in benzene molecule are sp^2 hybridised.

Choose the incorrect statement(s)

- (1) (a) and (c) only
- (2) (a) and (b) only
- (3) (b) and (c) only
- (4) (c) only

168. Kjeldahl method for estimation of nitrogen can be used to estimate the amount of nitrogen in which of the compound.

- (1) Aniline
- (2) Azobenzene
- (3) Pyridine
- (4) Nitrobenzene

169. The correct decreasing order of priority of the given functional groups is

- (1) $-\text{COOH} > -\text{COOR} > -\text{SO}_3\text{H} > -\text{CN}$
- (2) $-\text{COOH} > -\text{SO}_3\text{H} > -\text{COOR} > -\text{CN}$
- (3) $-\text{SO}_3\text{H} > -\text{COOH} > -\text{CN} > -\text{COOR}$
- (4) $-\text{SO}_3\text{H} > -\text{COOH} > -\text{COOR} > -\text{CN}$

170. Given below are two statements:

Statement I: Hydrolysis of cane sugar in acidic medium is a zero order reaction.

Statement II: Rate of reaction is temperature independent for zero order reaction.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

171. For a chemical reaction rate of reaction doubles on increasing temperature from 300 K to 310 K, then activation energy for the reaction will be

- (1) 18.4 kJ
- (2) 106.8 kJ
- (3) 24.44 kJ
- (4) 53.42 KJ

172. Match acids in List-I with their pKa values in List-II.

	List-I		List-II
a.	CF_3COOH	(i)	4.76
b.	$\text{C}_6\text{H}_5\text{COOH}$	(ii)	10
c.	CH_3COOH	(iii)	0.23
d.	$\text{C}_6\text{H}_5\text{OH}$	(iv)	4.19

Choose the correct answer from the options given below.

- (1) a(iii), b(iv), c(i), d(ii)
- (2) a(iii), b(iv), c(ii), d(i)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(ii), b(i), c(iv), d(iii)

173. Which among the following can be distinguished by iodoform test?

- (1) CH_3CHO and CH_3COCH_3
- (2) HCHO and $\text{C}_2\text{H}_5\text{CHO}$
- (3) CH_3COCH_3 and $\text{C}_2\text{H}_5\text{OH}$
- (4) CH_3CHO and $\text{C}_6\text{H}_5\text{CHO}$

174. Given below are two statements:

Statement I: Co gives violet colouration during borax bead test.

Statement II: Borax bead is H_3BO_3 .

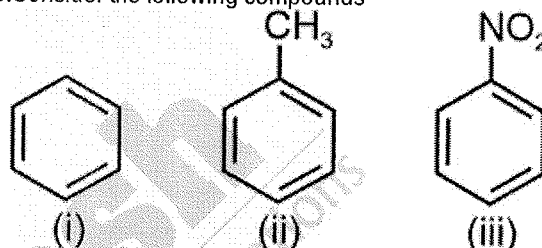
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

175. Ca^{2+} belongs to which of the following group of cations?

- (1) Group II
- (2) Group III
- (3) Group IV
- (4) Group V

176. Consider the following compounds



The correct order of rate of electrophilic substitution reaction in the given compounds is

- (1) (ii) > (i) > (iii)
- (2) (iii) > (ii) > (i)
- (3) (i) > (ii) > (iii)
- (4) (ii) > (iii) > (i)

177. Given below are two statements:

Statement I: Glucose is an aldohexose and is also known as dextrose.

Statement II: Glucose on prolonged heating with HI, forms n-hexane.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

178. The longest wavelength line in Lyman series of hydrogen atom spectrum is ($R_H = 109677 \text{ cm}^{-1}$)

- (1) 121.6 nm
- (2) 152.1 nm
- (3) 215.5 nm
- (4) 241.2 nm

179. Consider the following statements.

- (a) Each polypeptide in a protein has amino acids linked with each other in a specific sequence.
- (b) In β -pleated sheet structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds.
- (c) The main forces which stabilize 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages Van der Waals and electrostatic forces of attraction.

The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

CC-483

CC-483

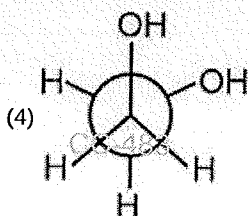
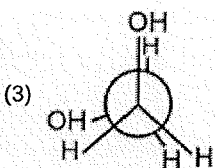
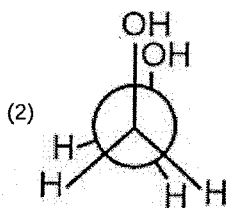
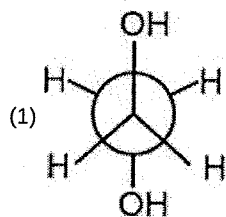
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180. Which of the following is the most stable conformation of ethylene glycol?



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25/04/2025

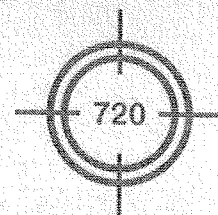


Aakash

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-II

Time : 180 Mins.

Answers

- | | | | | |
|---------|---------|----------|----------|----------|
| 1. (3) | 37. (3) | 73. (2) | 109. (1) | 145. (2) |
| 2. (2) | 38. (1) | 74. (2) | 110. (1) | 146. (3) |
| 3. (2) | 39. (2) | 75. (3) | 111. (3) | 147. (1) |
| 4. (3) | 40. (4) | 76. (1) | 112. (3) | 148. (1) |
| 5. (1) | 41. (1) | 77. (2) | 113. (2) | 149. (3) |
| 6. (1) | 42. (4) | 78. (3) | 114. (3) | 150. (4) |
| 7. (3) | 43. (3) | 79. (1) | 115. (2) | 151. (1) |
| 8. (1) | 44. (1) | 80. (3) | 116. (4) | 152. (3) |
| 9. (4) | 45. (4) | 81. (3) | 117. (2) | 153. (2) |
| 10. (2) | 46. (1) | 82. (4) | 118. (3) | 154. (2) |
| 11. (3) | 47. (3) | 83. (2) | 119. (4) | 155. (3) |
| 12. (3) | 48. (3) | 84. (4) | 120. (1) | 156. (2) |
| 13. (1) | 49. (3) | 85. (4) | 121. (1) | 157. (2) |
| 14. (4) | 50. (3) | 86. (3) | 122. (1) | 158. (1) |
| 15. (4) | 51. (4) | 87. (3) | 123. (1) | 159. (3) |
| 16. (4) | 52. (2) | 88. (4) | 124. (2) | 160. (4) |
| 17. (3) | 53. (3) | 89. (2) | 125. (1) | 161. (2) |
| 18. (4) | 54. (2) | 90. (1) | 126. (2) | 162. (2) |
| 19. (4) | 55. (3) | 91. (1) | 127. (1) | 163. (2) |
| 20. (3) | 56. (3) | 92. (1) | 128. (2) | 164. (1) |
| 21. (2) | 57. (4) | 93. (3) | 129. (1) | 165. (1) |
| 22. (4) | 58. (3) | 94. (2) | 130. (4) | 166. (4) |
| 23. (2) | 59. (2) | 95. (4) | 131. (4) | 167. (3) |
| 24. (3) | 60. (4) | 96. (3) | 132. (1) | 168. (1) |
| 25. (4) | 61. (4) | 97. (1) | 133. (1) | 169. (2) |
| 26. (3) | 62. (4) | 98. (2) | 134. (4) | 170. (2) |
| 27. (2) | 63. (4) | 99. (1) | 135. (1) | 171. (4) |
| 28. (2) | 64. (2) | 100. (2) | 136. (2) | 172. (1) |
| 29. (2) | 65. (2) | 101. (3) | 137. (2) | 173. (4) |
| 30. (3) | 66. (2) | 102. (3) | 138. (1) | 174. (2) |
| 31. (4) | 67. (1) | 103. (3) | 139. (2) | 175. (4) |
| 32. (4) | 68. (4) | 104. (3) | 140. (2) | 176. (1) |
| 33. (4) | 69. (2) | 105. (3) | 141. (2) | 177. (3) |
| 34. (1) | 70. (3) | 106. (2) | 142. (1) | 178. (1) |
| 35. (4) | 71. (4) | 107. (2) | 143. (4) | 179. (4) |
| 36. (3) | 72. (4) | 108. (1) | 144. (3) | 180. (4) |

Hints and Solutions

BIOLOGY

(1) Answer : (3)

Solution:

Flowers in mango are bisexual. Ethylene initiates flowering in mango.

(2) Answer : (2)

Solution:

Increased vacuolation is seen in the cells of elongation phase of the plant parts such as root tips.

(3) Answer : (2)

Solution:

Cells of tapetum possess dense cytoplasm and generally have more than one nucleus. They are polyploid.

(4) Answer : (3)

Solution:

A fruit which develops from other floral parts along with the development of ovary wall is called false fruit.

(5) Answer : (1)

Solution:

A typical angiosperm embryo sac shows the presence of 7-cells that are, one egg cell, two synergids, three antipodals and one central cell. But since, the central cell has two polar nuclei, the embryo sac is 8-nucleate and 7-celled at maturity.

(6) Answer : (1)

Solution:

Myotonic dystrophy is an example of autosomal dominant disorder.

(7) Answer : (3)

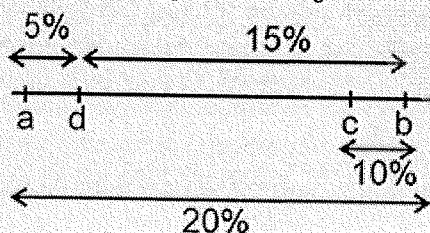
Solution:

Sickle-cell anaemia is an autosome linked recessive trait that is caused by the substitution of glutamic acid by valine at the sixth position of the β -globin chain of haemoglobin.

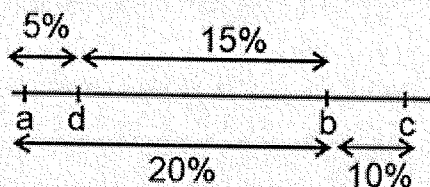
(8) Answer : (1)

Solution:

The possible arrangement of the genes can be



or



(9) Answer : (4)

Solution:

Dominance is not an autonomous feature of a gene. It depends much on the product of the gene and the production of a particular phenotype from this product.

(10) Answer : (2)

Solution:

In a eukaryotic cell, mRNA must be processed before it is translated. Polyadenylation, capping and splicing are the steps involved in mRNA processing.

(11) Answer : (3)

Solution:

The anticodon on tRNA for the codon 5' AGU 3' will be 5' ACU 3'

(12) Answer : (3)

Solution:

Promoter is a DNA sequence that provides binding site for RNA polymerase.

(13) Answer : (1)

Solution:

The mutation in operator region of operon leads to the constitutive expression of *lac* operon as the repressor cannot bind to the operator region.

(14) Answer : (4)

Solution:

In dicot roots, conjunctive tissue and pericycle are found.

(15) Answer : (4)

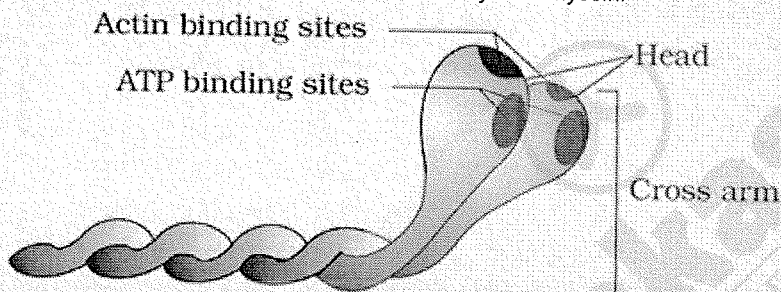
Solution:

Casparian strips are found in the endodermis of roots. Water molecules are unable to penetrate this layer in the areas of casparian strips.

(16) Answer : (4)

Solution:

A globular head with a short arm is called heavy meromyosin.



(17) Answer : (3)

Solution:

In a polymerase chain reaction, the number of DNA strands doubles with each cycle, following the formula:

Number of copies of DNA strands = 2^n .

Here 'n' is the number of cycles of PCR.

Thus, to generate 256 DNA strands from a single dsDNA piece, 8 PCR cycles are required.

(18) Answer : (4)

Solution:

EcoRI produces sticky ends by cutting the strand of DNA a little away from the centre of the palindromic sites but between the same two bases on opposite strands.

(19) Answer : (4)

Solution:

In all connective tissues, except blood and lymph, the cells secrete fibres of structural proteins called collagen or elastin.

(20) Answer : (3)

Solution:

Three major regions make up the brain stem, i.e., midbrain, pons and medulla oblongata.

Amygdala, hippocampus and hypothalamus are the parts of limbic system in forebrain.

(21) Answer : (2)

Solution:

The DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel.

The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid (autonomously replicating circular extra-chromosomal DNA) of *Salmonella typhimurium*.

(22) Answer : (4)

Solution:

The property common to both nerve cells (neurons) and muscle cells (myocytes) is excitability. Excitability refers to the ability of a cell to respond to stimuli by generating and propagating electrical impulses. Both neurons and muscle cells possess excitable membranes that can produce electrochemical impulses and conduct them along their surfaces.

The other properties listed- contractility, elasticity, and extensibility are primarily associated with muscle cells and are not the characteristics of nerve cells. Contractility is the ability of muscle cells to shorten and generate force. Elasticity refers to their ability to return to original length after stretching and extensibility is the capacity to be stretched without damage.

(23) Answer : (2)

Solution:

SAN is present in right upper corner of right atrium and AVN is present in lower left corner of right atrium. A thin, muscular wall called the inter-atrial septum separates right and left atrium. The atrium and the ventricle of the same side are also separated by a thick fibrous tissue called the atrio-ventricular septum.

(24) Answer : (3)

Solution:

Intercellular material of cartilage is solid and pliable and resists compression. Chondrocytes are enclosed in small cavities within matrix secreted by them.

(25) Answer : (4)

Solution:

Columba and *Pavo* belong to the class *Aves* so they have double circulation. *Ornithorhynchus* belongs to class *Mammalia*. It also has four-chambered heart and exhibits double circulation. *Carcharodon* belongs to the class *Chondrichthyes* and has two-chambered heart and exhibits single circulation.

(26) Answer : (3)

Solution:

Lateral meristems are generally not present from the very beginning of life of a plant and appear later than primary meristems. These are cylindrical meristems which result in increase in the girth of stems and roots. Apical meristems are found at the apices of stems and roots.

(27) Answer : (2)

Solution:

Nucleoid is the packaged structure of bacterial DNA. Clover-leaf is a secondary structure of tRNA. Regulator gene transcribes mRNA for protein synthesis.

(28) Answer : (2)

Solution:

Middle lamella is chiefly made up of calcium and magnesium pectate.

(29) Answer : (2)

Solution:

Plasmodesmata - Cytoplasmic connections
Mitochondria – Site for the synthesis of ATP
Axoneme - Core of cilia and flagella
Nucleolus - Site for rRNA synthesis

(30) Answer : (3)

Solution:

No organelle duplication takes place in the M phase of cell cycle.

(31) Answer : (4)

Solution:

Tetrad of cells is formed during telophase II.

(32) Answer : (4)

Solution:

Intermediate filaments form scaffolds for chromatin.

(33) Answer : (4)

Solution:

Mitosis helps in maintaining the cell size as it induces a cell to divide whenever the surface to volume ratio is disturbed, as the cell grows in size.

(34) Answer : (1)

Solution:

In both, green algae and brown algae, asexual reproduction is by flagellated zoospores. Red algae reproduce asexually by non-motile spores. Red algae are able to grow in great depth and they store food in the form of floridean starch and glycogen.

- (35) Answer : (4)
Solution:
 In *Marchantia*, male and female sex organs are present on different thalli, whereas in *Funaria* both male and female sex organs are present on the same thalli.
- (36) Answer : (3)
Solution:
Lactobacillus is used for the commercial production of lactic acid.
- (37) Answer : (3)
Solution:
 Biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate the prey population.
- (38) Answer : (1)
Solution:
 Chanda and Bastar areas are in Madhya Pradesh.
- (39) Answer : (2)
Solution:
 In co-extinction, when a species becomes extinct, the plant and animal species associated with it, in an obligatory way, also become extinct.
- (40) Answer : (4)
Solution:
 In the members of Phaeophyceae, the plant body is usually attached to the substratum by a holdfast and has a stalk, the stipe and leaf like photosynthetic organ, the frond. *Laminaria* belongs to Phaeophyceae.
- (41) Answer : (1)
Solution:
 Fibrinogens are needed for clotting or coagulation of blood. Globulins are primarily involved in defense mechanism of the body and the albumins help in osmotic balance. GLUT-4 enables glucose transport into cells.
- (42) Answer : (4)
Solution:
 The cutting of DNA at specific locations became possible with the discovery of the so-called 'molecular scissors'- restriction enzymes.
 The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase, which acts on cut DNA molecules and joins their ends.
- (43) Answer : (3)
Solution:
 Red muscle fibres have large amount of myoglobin and mitochondria while white muscle fibres have large amount of sarcoplasmic reticulum and depend on anaerobic process for energy.
- (44) Answer : (1)
Solution:
 In alveoli, where there is low $p\text{CO}_2$, high $p\text{O}_2$, lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin.
- (45) Answer : (4)
Solution:
 When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor e.g., Inhibition of succinic dehydrogenase by malonate which closely resembles the substrate succinate in the structure.
- (46) Answer : (1)
Solution:
 In peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.
- (47) Answer : (3)
Solution:
 The squamous epithelium is made of a single layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary. Squamous epithelium does not possess microvilli.
- (48) Answer : (3)
Solution:

Enzyme catalysts differ from inorganic catalysts in many ways. Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures (say above 40°C). However, enzymes isolated from organisms who normally live under extremely high temperatures (e.g., hot vents and sulphur springs), are stable and retain their catalytic power even at high temperatures (upto 80°-90°C).

(49) Answer : (3)

Solution:

Living organisms have a number of carbon compounds in which heterocyclic rings can be found. Some of these are nitrogenous bases - adenine, guanine, cytosine, uracil, and thymine.

(50) Answer : (3)

Solution:

During expiration, relaxation of the diaphragm and the external inter-costal muscles leads to an increase in intra-pulmonary pressure to slightly above the atmospheric pressure causing the expulsion of air from lungs.

(51) Answer : (4)

Solution:

Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testes in males and ovaries in females) are the organised endocrine bodies in our body.

(52) Answer : (2)

Solution:

Testes, ovaries and adrenal glands are paired structures.

Testes produce a group of hormones called androgens mainly testosterone.

Ovaries produce two steroid hormones called estrogen and progesterone.

Adrenal glands (adrenal cortex) release cortisol, androgenic steroids, mineralocorticoids, etc.

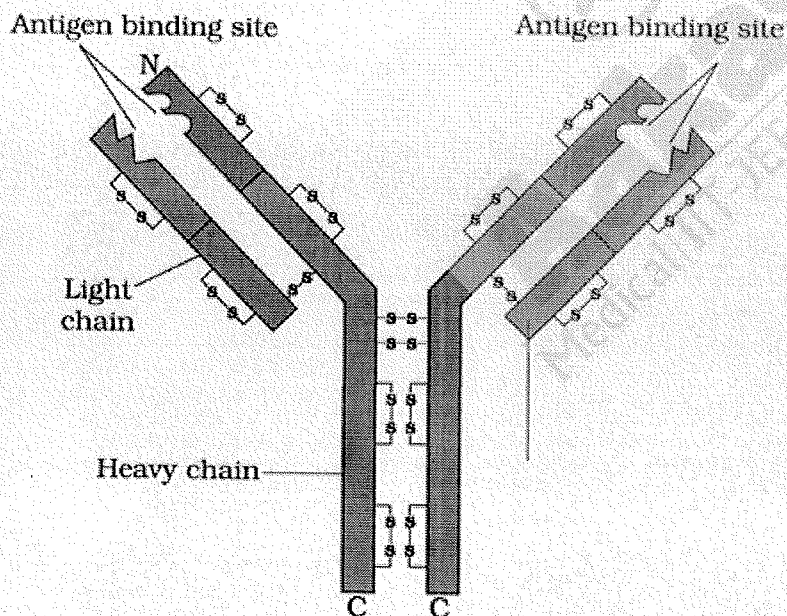
(53) Answer : (3)

Solution:

Here, 'X' refers to thymus which is a primary lymphoid organ, It is an unpaired gland and secretes peptide hormone called thymosin.

(54) Answer : (2)

Solution:



(55) Answer : (3)

Solution:

Active immunity is slow and takes time to give its full effective response.

Injecting the microbes deliberately during immunisation or infectious organisms gaining access into body during natural infection induce active immunity.

(56) Answer : (3)

Solution:

Acromegaly is caused due to hypersecretion of growth hormone during adulthood where as Graves' disease is caused due to hypersecretion of tetraiodothyronine (thyroxine).

(57) Answer : (4)

Solution:

Presence of notochord at some stage of their life is fundamental feature of all chordates. *Catla* is a fresh water bony fish which possesses air bladder that provides buoyancy. *Myxine* is a cyclostome in which scales are absent. Cycloid/ctenoid scales are present in *Catla*.

(58) Answer : (3)

Solution:

Under normal physiological conditions, the filtrate formed per day is 180 L and urine released per day is 1-1.5 L.

Hence, ratio between these two will be $\frac{180 \text{ L/day}}{1.5 \text{ L/day}} = 120 : 1$

(59) Answer : (2)

Solution:

Tyrannosaurus rex was 20 feet in height and had huge fearsome dagger-like teeth. *Pteranodon* was a flying reptile.

(60) Answer : (4)

Solution:

Even before Darwin, a French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. Wallace worked in Malay Archipelago. Hugo de Vries gave the idea of mutation.

(61) Answer : (4)

Solution:

In the given figure 'A' represents bobcat and 'B' represents wolf. Both bobcat and wolf are placental mammals and exhibits convergent evolution with tasmanian tiger cat and tasmanian wolf respectively.

(62) Answer : (4)

Solution:

Conditional reabsorption of Na^+ and water takes place in DCT. It is also capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions and NH_3 to maintain the pH and sodium-potassium balance in blood.

(63) Answer : (4)

Solution:

Limulus (King crab) possesses book gills for respiration.

(64) Answer : (2)

Solution:

Process in which isolated protoplasts from two different varieties of plants-each having desirable character can be fused to get hybrid plants is called somatic hybridisation.

(65) Answer : (2)

Solution:

Disarmed *Agrobacterium tumefaciens* vector is used to introduce nematode specific genes into the host plant to initiate RNAi.

(66) Answer : (2)

Solution:

Camembert cheese - *Penicillium camembertii*

Swiss Cheese - *Propionibacterium sharmanii*

Dosa - *Leuconostoc*

Pectinase - *Aspergillus niger*

(67) Answer : (1)

Solution:

In the given graph, curve 'a' is described by $\frac{dN}{dt} = rN$, and 'b' is described by $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

(68) Answer : (4)

Solution:

The special anatomy of leaves of the C_4 plant is called Kranz anatomy. Bell pepper is a C_3 plant. Both C_3 and C_4 plants utilize 2 molecules of NADPH per CO_2 fixed.

(69) Answer : (2)

Solution:

Non-cyclic photophosphorylation occurs in the membrane of granal thylakoids.

(70) Answer : (3)

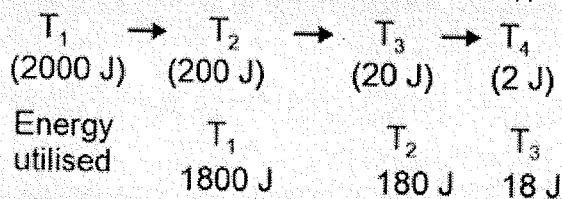
Solution:

In TCA cycle, the first decarboxylation reaction occurs during the conversion of oxalosuccinic acid to α -ketoglutaric acid.

(71) Answer : (4)

Solution:

According to 10% law, only 10% of the energy is transferred to each trophic level from the lower trophic level. Incident solar radiation energy 200,000 J. Only 1% of energy is trapped at T_1 level. Therefore, at T_1 (2000 J) of energy is present.



(72) Answer : (4)

Solution:

In alcoholic fermentation, acetaldehyde is reduced to alcohol by $NADH + H^+$ by the action of enzyme alcohol dehydrogenase.

(73) Answer : (2)

Solution:

The rate of decomposition is slower if the detritus is rich in lignin and chitin.

(74) Answer : (2)

Solution:

Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(75) Answer : (3)

Solution:

Neurospora is used extensively in biochemical and genetic work.

(76) Answer : (1)

Solution:

Imbricate aestivation is seen in *Cassia* and *Gulmohur*.

(77) Answer : (2)

Solution:

The flowers of Liliaceae family are bisexual and actinomorphic.

(78) Answer : (3)

Solution:

Sunflower and marigold show basal placentation.

(79) Answer : (1)

Solution:

In *Agaricus* and *Puccinia*, the mycelium is branched and septate.

(80) Answer : (3)

Solution:

Viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.

(81) Answer : (3)

Solution:

In ELISA, infection by pathogen can be detected by the presence of antigens (proteins, glycoproteins, etc.) or by detecting the antibodies synthesised against the pathogen.

(82) Answer : (4)

Solution:

Cuttlefish - Presence of radula

Prawn - Statocyst for balancing

Scorpion - Possesses book lungs for respiration

Roundworm - Dioecious in nature

(83) Answer : (2)

Solution:

In male frog, testes are found adhered to upper part of kidneys by a double fold of peritoneum called mesorchium. Endometrium is the innermost and myometrium is the middle layer present in the wall of human uterus.

(84) Answer : (4)

Solution:

The regions in between seminiferous tubules are called interstitial spaces which contain interstitial cells, immunologically competent cells and small blood vessels. Sertoli cells are present in the inner lining of seminiferous tubules.

(85) Answer : (4)

Solution:

Formation of first polar body occurs in tertiary follicle in ovaries. While second polar body is formed in fallopian tubes during fertilisation.

(86) Answer : (3)

Solution:

Prostate gland is an unpaired gland while epididymis is an accessory duct.

(87) Answer : (3)

Solution:

In male cockroaches, the genital pouch lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by 9th sternum.

(88) Answer : (4)

Solution:

The sperm head contains an elongated nucleus. The anterior portion of the head is covered by a cap-like structure, acrosome. A plasma membrane envelops the whole body of sperm.

(89) Answer : (2)

Solution:

Oral contraceptive pills containing progestogens alone or progesterone-estrogen in combination, inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms.

(90) Answer : (1)

Solution:

'Nirodh' is a popular brand of condom for the males. Use of condoms has increased in recent years due to its additional benefit of protecting the user from contracting STIs and AIDS.

Copper releasing IUDs - Multiload 375, Cu7

Hormone releasing IUDs - LNG-20

Copper released in copper releasing IUDs suppresses sperm motility and the fertilising capacity of sperms.

In tubectomy, a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through vagina, this technique is highly effective but its reversibility is very poor.

PHYSICS

(91) Answer : (1)

Solution:

A neutral body is charged positively when it loses electrons and hence the mass of the body will decrease.

(92) Answer : (1)

Solution:

$$E = -L \frac{di}{dt}$$

$$E = 0.02 \times \frac{4}{2 \times 10^{-2}}$$

$$= 4 \text{ V}$$

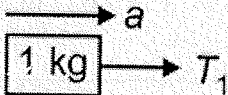
(93) Answer : (3)

Solution:

$$\text{Acceleration of system (a)} = \frac{F_{\text{net}}}{m}$$

$$= \frac{16}{8}$$

$$= 2 \text{ m/s}^2$$



$$T_1 = 1 \times 2$$

$$T_1 = 2 \text{ N}$$

(94) Answer : (2)

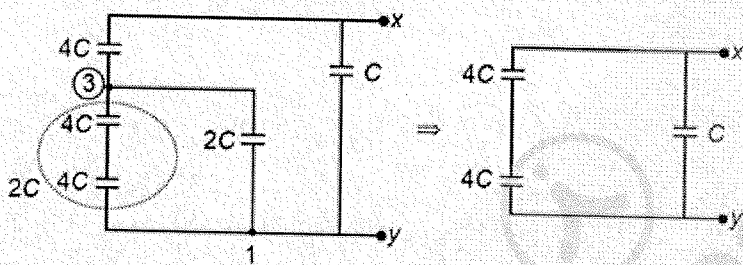
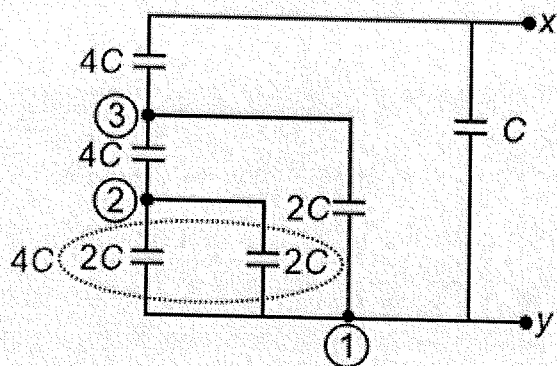
Solution:

$$V = \frac{kq}{r}$$

$$V_T = \frac{k}{1} [8 + 5 - 3 - 10] \times 10^{-6} = 0$$

(95) Answer : (4)

Solution:



$$C_{xy} = 2C + C = 3C$$

(96) Answer : (3)

Solution:

$$i = neAv_d \Rightarrow i = neA \times v \quad \dots(i)$$

$$2i = ne(2A) \times v' \quad \dots(ii)$$

$$v' = v$$

(97) Answer : (1)

Solution:

In half deflection method,

$$G = \frac{RS}{R+S}$$

$$= \frac{1000 \times 20}{1000 + 20}$$

$$= \frac{1000 \times 20}{980}$$

$$= \frac{1000}{49} \Omega$$

(98) Answer : (2)

Solution:

When p-side of diode is connected to higher potential, then diode is in forward biasing.

(99) Answer : (1)

Solution:

$$n_A = 10^{18} \text{ atoms/m}^3$$

$$n_i = 10^{16} \text{ m}^{-3}$$

$$n_e = \frac{n_i^2}{n_A} = \frac{10^{32}}{10^{18}} = 10^{14} \text{ m}^{-3}$$

(100) Answer : (2)

Solution:

$$\text{Energy of incident photon} = \frac{12400}{2000} = 6.2 \text{ eV}$$

$$\Rightarrow (K.E)_{\max} = h\nu - \phi$$

$$(K.E)_{\max} = 6.2 - 4.5$$

$$= 1.7 \text{ eV}$$

$$\therefore \text{Stopping potential} = -1.7 \text{ V}$$

(101) Answer : (3)

Solution:

$$\text{Energy of incident photon } E = 8 + 4.09$$

$$= 12.09 \text{ eV}$$

$$\Rightarrow \Delta E_{\text{photon}} = 12.09 \text{ eV}$$

$$= E_n - E_1$$

$$\therefore \frac{-13.6}{n^2} + 13.6 = 12.09$$

$$n = 3$$

(102) Answer : (3)

Solution:

$$\text{Mass defect } \Delta m = 0.2\% \text{ of } 1 \text{ kg}$$

$$= \frac{0.2}{100} \times 1$$

$$= 2 \times 10^{-3} \text{ kg}$$

$$\text{Energy released } E = \Delta mc^2$$

$$= 2 \times 10^{-3} \times (3 \times 10^8)^2$$

$$= 2 \times 10^{-3} \times 9 \times 10^{16}$$

$$= 18 \times 10^{13} \text{ J}$$

(103) Answer : (3)

Solution:

$$m = 1 \text{ kg}$$

$$W_{\text{net}} = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$= \frac{1}{2} m (\alpha^2 x_f^2 - \alpha^2 x_i^2)$$

$$= \frac{1}{2} \times (2700 - 100)$$

$$= 1300 \text{ J}$$

(104) Answer : (3)

Solution:

Here the motion is not accelerated, the resultant force parallel to the inclined plane must be zero.

$$\text{So, } F - mg \sin 30^\circ = 0$$

$$F = mg \sin 30^\circ$$

$$W = Fd \cos \theta$$

$$= mg \sin 30^\circ d \cos 0^\circ$$

$$= 500 \text{ J}$$

(105) Answer : (3)

Solution:

If multiple identical cells are connected in series in same sense, the total EMF increases.

(106) Answer : (2)

Solution:

$$F = \frac{v \Delta m}{\Delta t} = 20 \times 2 = 40 \text{ N}$$

$$a = \frac{F}{m} = \frac{40}{2} = 20 \text{ m/s}^2$$

(107) Answer : (2)

Solution:

$$v_{\text{avg}} = \frac{2x}{\frac{x}{2} + \frac{x}{5}}$$

$$= \frac{20}{7} \text{ m/s}$$

(108) Answer : (1)

Solution:

Speed at mid point is given by

$$v_m = \sqrt{\frac{u^2 + v^2}{2}}$$

$$v_m = \sqrt{\frac{100 + 400}{2}} = \sqrt{250} = 5\sqrt{10} \text{ m/s}$$

(109) Answer : (1)

Solution:

$$[MLT^{-2}] = [a]$$

$$\left[\frac{b}{L}\right] = [M^0 L^0 T^0]$$

$$[b] = [L]$$

(110) Answer : (1)

Solution:

$$V = a^3 = 8 \text{ cm}^3$$

$$\frac{\Delta V}{V} = \frac{3\Delta a}{a} \Rightarrow \Delta V = 3V \left(\frac{\Delta a}{a}\right) = 3 \times 8 \times \left(\frac{0.01}{2.00}\right)$$

$$= 0.12 \text{ cm}^3$$

$$V = (8 \pm 0.12) \text{ cm}^3$$

$$\text{Surface area} = 6a^2 = 6(2.00)^2 = 24.0 \text{ cm}^2$$

$$\frac{\Delta A}{A} = \frac{2\Delta a}{a} \Rightarrow \Delta A = 2A \left(\frac{\Delta a}{a}\right) = 0.24$$

$$\therefore A = (24.0 \pm 0.24) \text{ cm}^2$$

(111) Answer : (3)

Solution:

$$x = 10 \cos \left(2\pi(1) + \frac{\pi}{4} \right)$$

$$= 10 \cos \left(2\pi + \frac{\pi}{4} \right)$$

$$= 10 \cos \frac{\pi}{4}$$

$$= 10 \times \frac{1}{\sqrt{2}} = 5\sqrt{2} \text{ cm}$$

(112) Answer : (3)

Solution:

$$\text{Velocity of sound } v = \sqrt{\frac{\gamma P}{\rho}}$$

$\rho_{\text{moist air}} < \rho_{\text{dry air}}$

Velocity of sound in moist air is more than in dry air.

(113) Answer : (2)

Solution:

For first minimum, path difference $a \sin \theta = \lambda$

$$a \sin 30^\circ = \lambda$$

$$a = 2 \times \lambda = 2\lambda$$

For first secondary maximum, path difference

$$a \sin \theta' = \frac{3}{2} \lambda$$

$$\sin \theta' = \frac{3\lambda}{2(2\lambda)}$$

$$\theta' = \sin^{-1} \left(\frac{3}{4} \right)$$

(114) Answer : (3)

Solution:

Infrared rays are used in remote switches, microwaves are used in cooking, and UV rays are used to kill germs in water purifiers.

Gamma rays are used in medicine to destroy cancer cells.

(115) Answer : (2)

Solution:

$$\Delta l = \frac{FL}{AY}$$

$$= \frac{(100 \times 10^3 \text{ N}) \times 1 \text{ m}}{3.14 \times (10^{-2} \text{ m})^2 \times 2 \times 10^{11} \text{ N/m}^2}$$

$$= 1.59 \text{ mm}$$

(116) Answer : (4)

Solution:

Velocity of object A with respect to object B

$$\begin{aligned}\vec{v}_{A/B} &= \vec{v}_A - \vec{v}_B \\ &= 3\hat{i} + 4\hat{j} - (7\hat{i} - 3\hat{j}) \\ &= 3\hat{i} + 4\hat{j} - 7\hat{i} + 3\hat{j} \\ &= (-4\hat{i} + 7\hat{j}) \text{ m/s}\end{aligned}$$

(117) Answer : (2)

Solution:

Centre of mass of this rigid rod, $x_{cm} = \frac{\int x dm}{\int dm}$

where $\frac{dm}{dx} = 3x \Rightarrow dm = 3x dx$

$$\therefore \frac{\int_0^1 3x^2 dx}{\int_0^1 3x dx} = \frac{3 \times \frac{x^3}{3} \Big|_0^1}{3 \times \frac{x^2}{2} \Big|_0^1} = \frac{3 \times \frac{1}{3}}{3 \times \frac{1}{2}} = \frac{2}{3} \text{ m}$$

(118) Answer : (3)

Solution:

In a perfectly rigid body, there is no internal motion. The work done by external torques is therefore, not dissipated and goes on to increase the rotational kinetic energy of the body.

Torque $\tau = I\alpha \Rightarrow \alpha$ is proportional to $\frac{1}{I}$

(119) Answer : (4)

Solution:

The two masses will always remain diametrically opposite to keep revolution around a common centre.



$$U = \frac{-2GM^2}{R} - \frac{GM^2}{2R} = \frac{-5GM^2}{2R}$$

(120) Answer : (1)

Solution:

Physical balance measures the mass whereas spring balance measures the weight.

Reading on spring balance increases as $g_{\text{pole}} > g_{\text{equator}}$

Reading on physical balance remains the same.

(121) Answer : (1)

Solution:

For green light, $\sin C = \sin \theta = \frac{1}{\mu} = \frac{3}{4}$

Yellow, orange and red have higher $\lambda \Rightarrow$ lower $\mu \Rightarrow$ higher critical angle.

$\therefore$ These will be in emergent ray.

(122) Answer : (1)

Solution:

$$\frac{1}{v} - \frac{1.5}{-120} = \frac{1-1.5}{40} = \frac{-1}{80} \Rightarrow \frac{1}{v} + \frac{1}{80} = \frac{-1}{80}$$

$$\Rightarrow \frac{1}{v} = \frac{-2}{80} = \frac{-1}{40} \Rightarrow v = -40 \text{ cm}$$

(123) Answer : (1)

Solution:

Gauss law is valid in uniform electric field. If we make a gaussian surface in a uniform electric field, then net flux through it will be zero and hence net charge inside is zero.

(124) Answer : (2)

Solution:

By Lenz's law, to decrease the magnetic flux current will decrease.

(125) Answer : (1)

Solution:

Due to symmetry, $M' = \frac{M}{4}$

(126) Answer : (2)

Solution:

$$M = NIA$$

$$= I\pi R^2$$

$$\tau = \left| \vec{M} \times \vec{B} \right|$$

$$= MB \sin 60^\circ$$

$$= \frac{\pi R^2 IB \sqrt{3}}{2}$$

(127) Answer : (1)

Solution:

$$\vec{F} = q \left(\vec{v} \times \vec{B} \right)$$

$$\vec{F} \neq 0$$

So, the charge's velocity, path and momentum will change.

(128) Answer : (2)

Solution:

$$\omega^2 = \frac{1}{LC}$$

$$10^4 = \frac{1}{\frac{1}{2} \times 10^2 \times 10^{-3} \times C}$$

$$C = \frac{2}{10^6 \times 10^{-3}}$$

$$C = 2 \text{ mF}$$

(129) Answer : (1)

Solution:

$$\frac{N_P}{N_S} = \frac{I_S}{I_P} = \frac{1}{3}$$

(130) Answer : (4)

Solution:

We know that, $h = \frac{2T \cos \theta}{R \rho g}$, if $\theta = 0^\circ$ then liquid will rise. Liquid will not rise or fall when $\theta = 90^\circ$.

Venturimeter is based on Bernoulli's principle.

(131) Answer : (4)

Solution:

Since the ball is moving with constant velocity, hence the net force on ball would be zero.

(132) Answer : (1)

Solution:

$$m \times s \times \Delta T = M \times L + 10 \times M$$

$$\Rightarrow 100 \times 1 \times 60 = M \times 540 + 10 \times M$$

$$\Rightarrow M = \frac{100 \times 60}{550}$$

$$\Rightarrow M = \frac{120}{11} = 10.9 \text{ g}$$

(133) Answer : (1)

Solution:

From first law of thermodynamics we know that $Q = \Delta U + W$ and if $Q = 0 \Rightarrow \Delta U + W = 0$

$\Rightarrow \Delta U = -W \Rightarrow$ process should be adiabatic.

(134) Answer : (4)

Solution:

Since, process is isobaric, therefore $V \propto T$

$$W = nR\Delta T \Rightarrow W = 3 \times 2 \times (3T - T)$$

$$= 3 \times 2 \times 2 \times [273 + 47]$$

$$= 3 \times 4 \times 320$$

$$= 3840 \text{ cal}$$

(135) Answer : (1)

Solution:

For Argon, $K.E = \frac{3}{2} k_B T$

For Oxygen, $K.E = \frac{5}{2} k_B T$

$\Rightarrow \text{Ratio} = \frac{3}{5}$

CHEMISTRY

(136) Answer : (2)

Solution:

Solution which show large negative deviation from Raoult's law form maximum boiling azeotrope.

(137) Answer : (2)

Solution:

$\text{NaCl} \rightarrow i = 1 + \alpha = 1 + 0.9 = 1.9$

$\text{CaCl}_2 \rightarrow i = 1 + 2\alpha = 1 + 2 \times 0.6 = 2.2$

$\text{AlCl}_3 \rightarrow i = 1 + 3\alpha = 1 + 3 \times 0.5 = 2.5$

$\text{K}_2[\text{Fe}(\text{CN})_6] \rightarrow i = 1 + 3\alpha = 1 + 3 \times 0.7 = 3.25$

(138) Answer : (1)

Solution:

Covalent nature of ionic bond increases with increase in charge on both cation and anion and with decrease in size of cation and increase in size of anion

(139) Answer : (2)

Solution:

$\text{I}_3^- \rightarrow \text{Linear (planar)}$

$\text{ClF}_3 \rightarrow \text{Bent T shape (planar)}$

$\text{BF}_3 \rightarrow \text{Trigonal planar}$

$\text{XeO}_3 \rightarrow \text{Pyramidal}$

$\text{B}_2\text{H}_6 \rightarrow \text{non-planar}$

(140) Answer : (2)

Solution:

$\text{O}_2 = 2$

$\text{O}_2^+ = 2.5$

$\text{O}_2^- = 1.5$

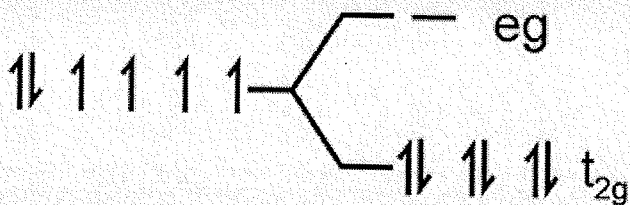
$\text{O}_2^{2-} = 1$

(141) Answer : (2)

Solution:

CN^- ligand act as strong field ligand.

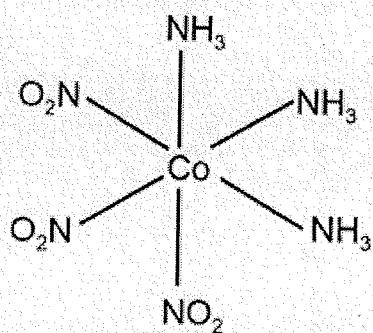
$\text{Co}^{3+} \rightarrow [\text{Ar}]4s^0 3d^6$



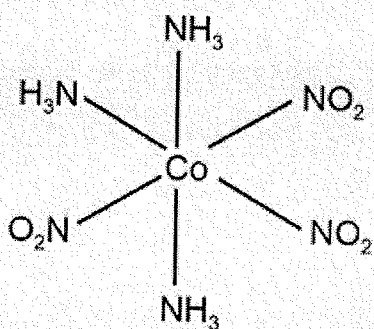
So correct configuration is $t_{2g}^6 eg^0$

(142) Answer : (1)

Solution:



Fac –

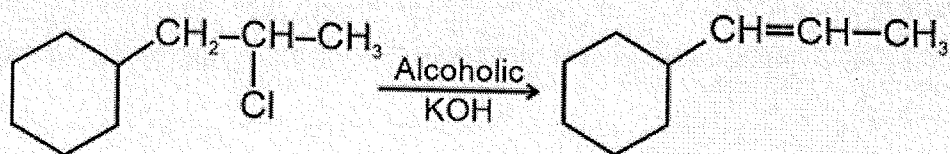


Mer –

• $[\text{Co(en)}_3]^{3+}$ complex show optical isomerism.

(143) Answer : (4)

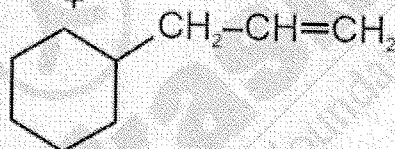
Solution:



(A)

Major

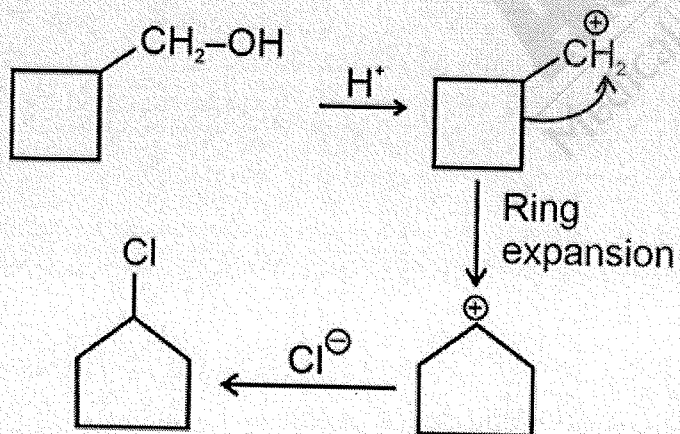
+



(Minor)

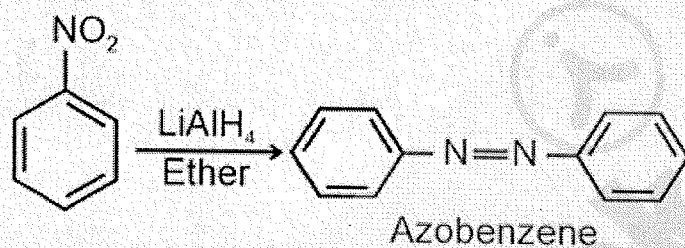
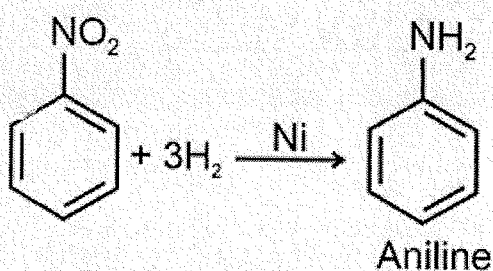
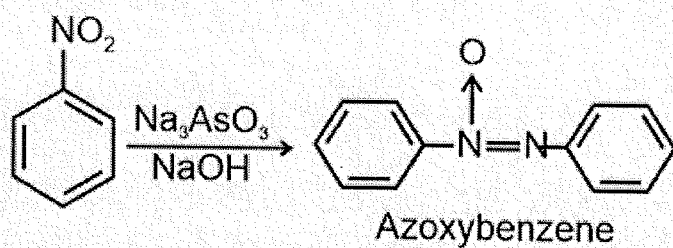
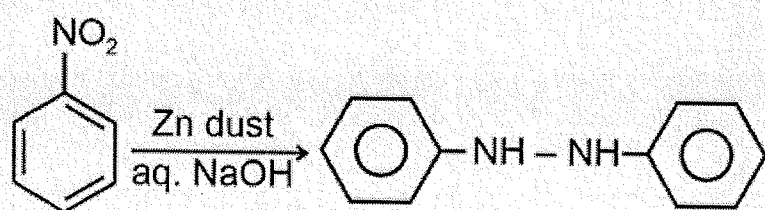
(144) Answer : (3)

Solution:



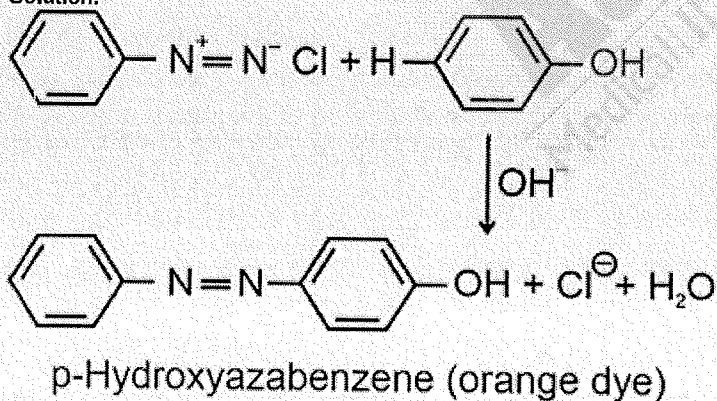
(145) Answer : (2)

Solution:



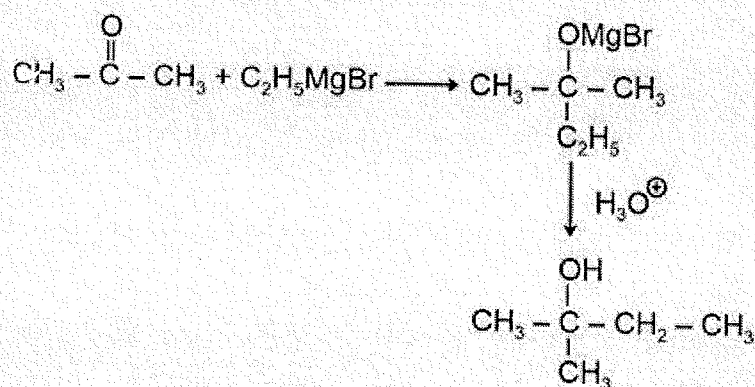
(146) Answer : (3)

Solution:



(147) Answer : (1)

Solution:



(148) Answer : (1)

Solution:

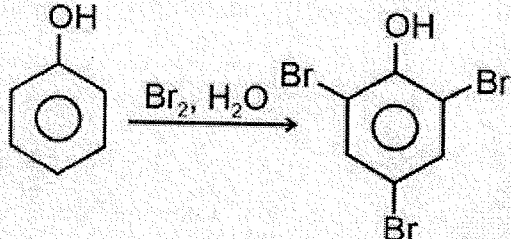
$$\lambda = \frac{h}{mv}$$

$$\lambda \propto \frac{1}{m}$$

Smaller is the molecular mass, longer is the wavelength.

(149) Answer : (3)

Solution:



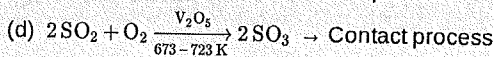
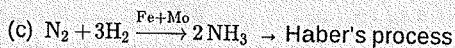
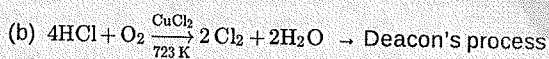
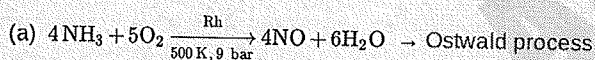
(150) Answer : (4)

Solution:

Primary alcohol $\xrightarrow{\text{KMnO}_4, \text{H}^+}$ Carboxylic acid.

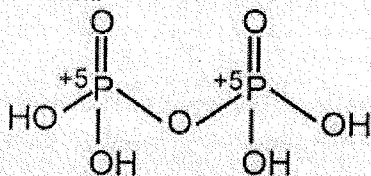
(151) Answer : (1)

Solution:



(152) Answer : (3)

Solution:

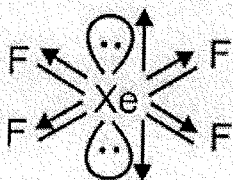


Number of P—OH bonds in $\text{H}_4\text{P}_2\text{O}_6 = 4$

(153) Answer : (2)

Solution:

XeF_4 has zero dipole moment. It has Square planar structure due to which the bond moments of Xe—F cancel each other



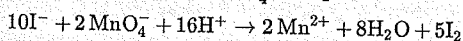
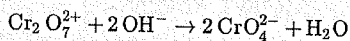
(154) Answer : (2)

Solution:

La^{3+} and Lu^{3+} are diamagnetic species. Thorium exhibits only +4 oxidation state.

(155) Answer : (3)

Solution:



(156) Answer : (2)

Solution:

$$\Delta H(\text{kJ mol}^{-1})$$

$$\left(\frac{1}{2}\text{A} \rightarrow \text{B}\right) \times 2 \Delta H = -200$$

$$3\text{B} \rightarrow 2\text{C} + \text{D} \quad \Delta H = +120$$

$$2\text{D} \rightarrow \text{E} + \text{A} \quad \Delta H = +325$$

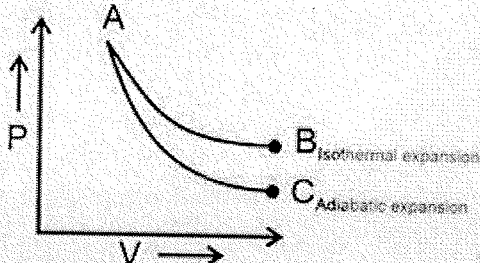
$$\Delta H \text{ for } \text{B} + \text{D} \rightarrow \text{E} + 2\text{C}$$

$$= (-200 + 120 + 325) \text{ kJ mol}^{-1}$$

$$= +245 \text{ kJ mol}^{-1}$$

(157) Answer : (2)

Solution:



During adiabatic expansion, cooling takes place.

(158) Answer : (1)

Solution:

Since density of water 1 g/mL

$\therefore$ mass of water in 36 mL = 36 g

Moles = 2 moles

(159) Answer : (3)

Solution:

(Atomic number)	(Symbol)
102	Unb
115	Uup
114	Uuq
107	Uns

(160) Answer : (4)

Solution:

Higher is the reduction potential, greater is the oxidising power,
So, order of reducing power will be : $\text{C} > \text{B} > \text{A}$

(161) Answer : (2)

Solution:

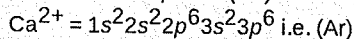
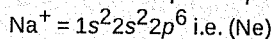
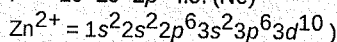
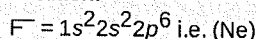
$$\Delta G = -RT \ln K_C$$

$$= -8.314 \times 300 \times \ln(2 \times 10^{13})$$

$$\Delta G = -7.64 \times 10^4 \text{ J mol}^{-1}$$

(162) Answer : (2)

Solution:



(163) Answer : (2)

Solution:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$$

$$\alpha = \frac{10}{250} = 0.04$$

As α is quite less; $\alpha \ll 1$

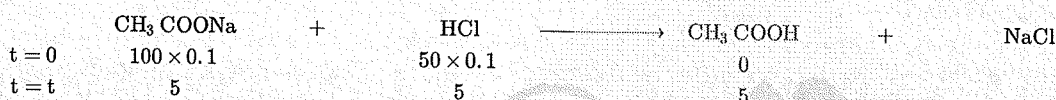
Hence, it is negligible

$$K_a = c\alpha^2 = 0.02(0.04)^2$$

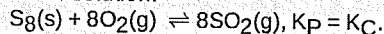
$$= 3.2 \times 10^{-5}$$

(164) Answer : (1)

Solution:



Now solution consists of CH_3COOH and CH_3COONa i.e. weak acid and salt of weak acid and strong base form acidic buffer solution.



(165) Answer : (1)

Solution:

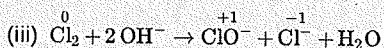
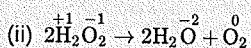
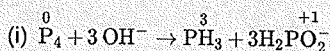
Order of first ionisation enthalpy :

$\text{Si} > \text{Ge} > \text{Pb} > \text{Sn}$

(166) Answer : (4)

Solution:

The reaction in which same element gets oxidised as well as reduced is known as disproportionation reaction.

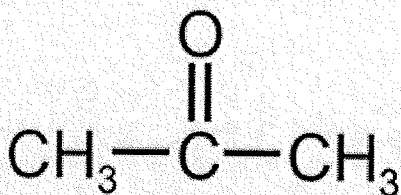


All are disproportionation reactions

(167) Answer : (3)

Solution:

$\text{C}_3\text{H}_6\text{O}$: a.

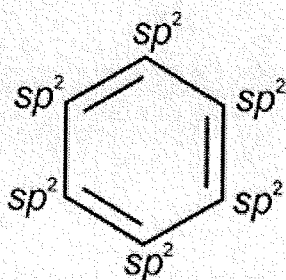


cannot show metamerism.

b. $\text{H}_3\text{C} - \overset{\oplus}{\text{CH}_2}$ has 3α hydrogen and it is stabilised by hyperconjugation while in $\overset{\oplus}{\text{CH}_3}$ there is no possibility of hyperconjugation.

c. Hyperconjugation is a permanent effect.

d.

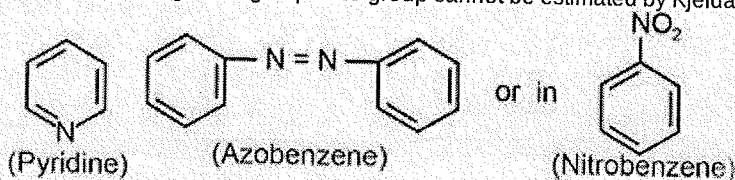


All the carbon atoms are sp^2 hybridised.

(168) Answer : (1)

Solution:

Nitrogen in a ring / nitro group / azo group cannot be estimated by Kjeldahl method.



Nitrogen cannot be estimated by Kjeldahl method

(169) Answer : (2)

Solution:

The correct priority order is

$-\text{COOH} > -\text{SO}_3\text{H} > -\text{COOR} > -\text{CN}$

(170) Answer : (2)

Solution:

Hydrolysis of cane sugar in acidic medium is a pseudo first order reaction.

(171) Answer : (4)

Solution:

$$\log \frac{k_1}{k_2} = \frac{E_a}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log 2 = \frac{E_a}{2.303 \times 8.314} \left(\frac{1}{300} - \frac{1}{310} \right)$$

$$E_a = 0.3 \times 2.303 \times 8.314 \times 300 \times 31 \text{ J}$$

$$= 53.42 \text{ kJ}$$

(172) Answer : (1)

Solution:

Acid	pKa
CF_3COOH	0.23
$\text{C}_6\text{H}_5\text{COOH}$	4.19
CH_3COOH	4.76
$\text{C}_6\text{H}_5\text{COOH}$	10

(173) Answer : (4)

Solution:

CH_3CHO will give yellow precipitate with $\text{I}_2 + \text{NaOH}$ but $\text{C}_6\text{H}_5\text{CHO}$ will not give yellow precipitate.

(174) Answer : (2)

Solution:

Borax bead is $\text{NaBO}_2 + \text{B}_2\text{O}_3$.

(175) Answer : (4)

Solution:

Na^+ , Mg^{2+} and Ca^{2+} belongs to group V cations.

(176) Answer : (1)

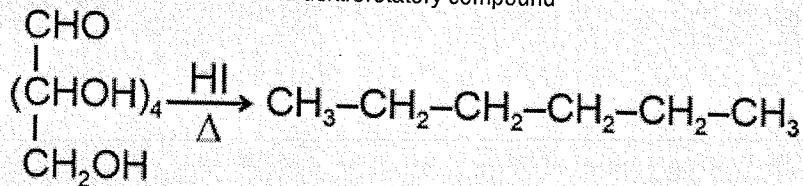
Solution:

Electron donating group attached with benzene ring increases the rate of electrophilic substitution reaction.

(177) Answer : (3)

Solution:

Glucose in an aldohexose and dextrorotatory compound



(178) Answer : (1)

Solution:

$$\text{Sol.: } \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Rightarrow \frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\Rightarrow \frac{1}{\lambda} = R_H \left(\frac{4-1}{4} \right)$$

$$\Rightarrow \lambda = \frac{4}{3R_H} = \frac{4}{3 \times 109677} \text{ cm}$$

$$\Rightarrow \lambda = 1.2157 \times 10^{-5} \text{ cm} = 1.2157 \times 10^{-7} \text{ m}$$

$$\Rightarrow \lambda = 1.2157 \times 10^{-9} \text{ m} \approx 121.6 \text{ nm}$$

(179) Answer : (4)

Solution:

The secondary structure of protein refers to the shape in which a long polypeptide chain each exist. They are found to exist in two different types of structures viz. α -helix and β -pleated sheet structure.

(180) Answer : (4)

Solution:

